

Unifying Special Relativity and Quantum Mechanics

Lecture 30 — The Non-Abelian Higgs Mechanism, and a Standard Model Primer

(completing the equivalence-theorem check; the general $SU(N)$ construction and the Gram mass matrix $M_{\alpha\beta}^2 = 2g^2\Phi_0^\dagger X^\alpha X^\beta \Phi_0$; two worked examples — $SU(2) \rightarrow 1$ with three degenerate W 's and Higgs unitarisation of $W_L W_L$ scattering, and the Georgi–Glashow model $SU(2) \rightarrow U(1)$ with $W^\pm$ of charge ± 1 ; then: the Standard Model gauge group, the doublet $H \sim (1, 2, \frac{1}{2})$, and $Q = T^3 + Y$)

Typed version of the handwritten notes, with added explanations in blue boxes

Added explanation (not in the original notes) — How to read this document

Conventions. Exactly as in Lectures 1–29: normal text and the yellow / pink / grey / green boxes are a faithful transcription of the handwritten sheets, and the text colours follow the pens of the original (blue, magenta, green, red, violet, pencil, brown). Everything inside a blue box like this one is material I have added; it is not in the original notes.

Pagination — this is the longest scan so far, so read this carefully.

1. **The first two sheets are a repeat.** This scan opens at sheet 7 of the *Higgs mechanism* chapter, but Lecture 29 already ran [p. 1*****]–[p. 8*****]. Sheets 7 and 8 are duplicated from the end of that scan. Section 1 records their content compactly, clearly marked; for the full transcription see Lecture 29, Section 9.
2. **Sheet 9 is the one you have been waiting for.** It completes the calculation that Lecture 29 broke off in the middle of — the ultra-relativistic limit of the $A_L A_L$ amplitude — and states the punchline. Section 2.
3. **Sheets 10–29 are the body of the lecture:** the non-abelian Higgs mechanism in general, then two fully worked examples. All of these carry *fourteen* stars, [p. 10*****]–[p. 29*****], because they belong to the same chapter.
4. **A new chapter begins after a blank page.** PDF page 24 is blank (the back of a sheet). Then the numbering restarts at 1 for *A Standard Model primer*. That is the *fifteenth* restart of the notes, so those sheets carry *fifteen* stars: [p. 1*****]–[p. 6*****]. Fourteen stars means the Higgs chapter; fifteen means the Standard Model chapter. Count them if in doubt.

So: PDF page n is Higgs sheet $n + 6$ for $n \leq 23$; page 24 is blank; and PDF page n is Standard Model sheet $n - 24$ for $n \geq 25$. Twenty-nine sheets of content, nothing missing, nothing out of order.

It stops mid-development, as usual. The last sheet, [p. 6*****], expands $(D_\mu H)^\dagger (D^\mu H)$ for the Standard Model and isolates the terms quadratic in the gauge fields — including, crucially, a $W_\mu^3 B^\mu$ cross term. It breaks off exactly before that 2×2 mass matrix is diagonalised. Section 18 does the diagonalisation: the photon, the Z , the weak mixing angle, and $\rho = 1$.

Slips and reading notes, flagged in place.

1. On [p. 10*****] the two numbered steps read “1)” and “3)”; the second is clearly “2)”, since [p. 11*****] then continues with 3) and 4). Renumbered here.
2. On [p. 10*****] the quartic in the potential is written with an ambiguous exponent; it is $\lambda(\Phi^\dagger \Phi)^2$, as [p. 11*****] and every later sheet confirm.
3. On [p. 18*****] the factors of $\frac{1}{2}$ drift between σ^A and $\sigma^A/2$ inside the cross-term bracket. The conclusion (the bracket vanishes) is unaffected; see the blue box there.

4. On [p. 28*****] the rotation matrix is written with $+\sin\theta$ in the upper-right entry; the result quoted immediately below it, $\cos\theta W_\mu^1 - \sin\theta W_\mu^2$, requires $-\sin\theta$ there. Corrected above, matching the result.
5. On [p. 20*****] the vertex factor is quoted as $\frac{iM_W^2}{v} 2\delta_{AB}g^{\mu\nu}$; this is correct with $M_W^2 = \frac{1}{4}g^2v^2$ — see the blue box for the check.
6. Throughout the Georgi–Glashow section the adjoint index and the “vector in field space” index are both written $A, B, \dots$ or $a, b, \dots$ interchangeably. They are the same index — in the adjoint representation the field *is* a vector in the algebra.

Notation. T^A generic generators; X^α broken, T^a unbroken (the CCWZ convention of Lectures 27–29 is kept). G_μ^A generic gauge fields; W_μ^A the $SU(2)$ gauge fields; B_μ the hypercharge field; G_μ^A (Standard Model chapter) the gluons. H a complex $SU(2)$ doublet; Φ either a generic fundamental ([p. 10*****]–[p. 13*****]) or a real adjoint triplet ([p. 22*****]–[p. 29*****]), stated in each case. $v^2 = \mu^2/\lambda$ throughout, and σ^A are the Pauli matrices.

Added explanation (not in the original notes) — Where we left off — recap of Lecture 29

Lecture 29 did the *abelian* Higgs mechanism completely. The chain was:

- Take scalar QED with a wrong-sign mass, $\mathcal{L} = |D_\mu\Phi|^2 + \mu^2|\Phi|^2 - \lambda|\Phi|^4 - \frac{1}{4}F^2$, gauge invariant under $\Phi \rightarrow e^{-i\alpha(x)}\Phi$, $A_\mu \rightarrow A_\mu + \frac{1}{g}\partial_\mu\alpha$. The vacuum sits at $|\Phi|^2 = v^2/2$.
- Write $\Phi = \frac{1}{\sqrt{2}}(v + h)e^{i\pi/v}$ and use the *arbitrariness* of $\alpha(x)$ to choose $\alpha = \pi/v$ — the **unitary gauge**. The Goldstone cancels identically out of Φ .
- Substituting gives a massive Higgs, $m_h^2 = 2\mu^2$, and a **Proca** term $\frac{1}{2}g^2v^2A_\mu A^\mu$, i.e. $m_A^2 = g^2v^2$. The Goldstone has been **eaten**: DOF counting $2 + 2 = 4$ before, $1 + 3 = 4$ after.
- Where did it go? Into the longitudinal polarisation. The check — tree-level $A_L A_L \rightarrow A_L A_L$ through s, t, u Higgs exchange — was set up in full and the scan cut off one substitution short of the answer.

This scan supplies that substitution on [p. 9***], and the answer is exactly the four-Goldstone amplitude. From [p. 10*****] onwards everything is new.**

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PART I — Closing the abelian chapter [p. 7***]–[p. 9*****]**

1 The duplicated sheets

[p. 7*****]–[p. 8*****]

Added explanation (not in the original notes) — These two sheets already appeared at the end of Lecture 29

Sheets [p. 7*****] and [p. 8*****] are physically the same pages as the last two of the Lecture 29 scan, re-scanned here. Their content is transcribed in full in Lecture 29, Section 9. For continuity, here is what they contain, compactly.

Sheet [p. 7***].** The conclusion of argument 1): the eaten Goldstone appears as the longitudinal DOF in the ultrarelativistic limit. Then the exercise is set up: the interaction Lagrangian $\mathcal{L}_{\text{int}} = \frac{g^2}{2}(h^2 + 2hv)A_\mu A^\mu$ (unitary-gauge fields, the “u” subscript omitted), the process $A_L(p_1) + A_L(p_2) \rightarrow A_L(p_3) + A_L(p_4)$, and the two Feynman rules

$$\mathcal{L}_{\text{int}}^{(3)} = g^2 hv g_{\mu\nu} A^\mu A^\nu \Rightarrow 2ivg^2 g_{\mu\nu}, \quad \mathcal{L}_{\text{int}}^{(4)} = \frac{g^2}{2} h^2 g_{\mu\nu} A^\mu A^\nu \Rightarrow 2ig^2 g_{\mu\nu}.$$

Sheet [p. 8***].** The three tree diagrams (s, t, u channel Higgs exchange), the s -channel amplitude, and the full amplitude

$$i\mathcal{M} = (-i)4v^2 g^4 \varepsilon_L^\mu(p_1) \varepsilon_L^\nu(p_2) \varepsilon_L^\rho(p_3)^* \varepsilon_L^\sigma(p_4)^* \left(\frac{g_{\mu\nu} g_{\rho\sigma}}{s - m_h^2} + \frac{g_{\mu\rho} g_{\nu\sigma}}{t - m_h^2} + \frac{g_{\mu\sigma} g_{\nu\rho}}{u - m_h^2} \right),$$

followed by the energy-ladder sketch and the statement that we work at $s, t, u \gg m_A^2$ but not necessarily below the Higgs mass, so the UR limit is used for the gauge boson.

The next sheet completes the substitution, and that is where the new material begins.

2 The ultra-relativistic limit, and the equivalence theorem

[p. 9*****]

Using $\varepsilon_L^\mu(p) \simeq \frac{p^\mu}{M} = \frac{p^\mu}{gv}$

[p. 9*****]

$$\begin{aligned} i\mathcal{M} &\simeq (-i)4v^2 g^4 \left(\frac{1}{gv} \right)^4 \left(\frac{(p_1 \cdot p_2)(p_3 \cdot p_4)}{s - m_h^2} + \frac{(p_1 \cdot p_3)(p_2 \cdot p_4)}{t - m_h^2} + \frac{(p_1 \cdot p_4)(p_2 \cdot p_3)}{u - m_h^2} \right) \\ &= \frac{(-i)}{v^2} \left[\frac{(2p_1 \cdot p_2)(2p_3 \cdot p_4)}{s - m_h^2} + \frac{(2p_1 \cdot p_3)(2p_2 \cdot p_4)}{t - m_h^2} + \frac{(2p_1 \cdot p_4)(2p_2 \cdot p_3)}{u - m_h^2} \right] \end{aligned}$$

Since in the UR limit for the gauge boson the external particles of this process are massless, we have

$$\begin{cases} s = 2p_1 \cdot p_2 = 2p_3 \cdot p_4 \\ t = -2p_1 \cdot p_3 = -2p_2 \cdot p_4 \\ u = -2p_1 \cdot p_4 = -2p_2 \cdot p_3 \end{cases}$$

We get

$$i\mathcal{M} \simeq \frac{(-i)}{v^2} \left(\frac{s^2}{s - m_h^2} + \frac{t^2}{t - m_h^2} + \frac{u^2}{u - m_h^2} \right)$$

↔ This is **precisely the same amplitude** that we found when we computed the scattering of Goldstone bosons in the theory with spontaneous symmetry breaking $SO(2)$

This result enforces the interpretation of Goldstone bosons as longitudinal polarisations of massive gauge fields in the broken phase

Added explanation (not in the original notes) — The two sign traps in this one line of algebra

Why the 2's appear. The prefactor is $4v^2g^4 \cdot (gv)^{-4} = 4/v^2$, and the 4 is then distributed as $4(p_1 \cdot p_2)(p_3 \cdot p_4) = (2p_1 \cdot p_2)(2p_3 \cdot p_4)$ so that each bracket is directly a Mandelstam variable. That is the only reason for the rewriting.

Why t and u come with minus signs and still give $+t^2, +u^2$. For massless external legs $t = (p_1 - p_3)^2 = -2p_1 \cdot p_3$, so $2p_1 \cdot p_3 = -t$. The t -channel numerator is a *product* of two such factors, $(2p_1 \cdot p_3)(2p_2 \cdot p_4) = (-t)(-t) = +t^2$. The two minus signs cancel. Same for u . This is why all three terms enter with the same sign, despite s being “+” and t, u being “−”.

And why g vanishes entirely. The gauge coupling appears squared in each vertex ($2ivg^2$) and inversely to the fourth power through the four polarisation vectors ($\varepsilon_L \sim p/gv$). The two cancel exactly. The final amplitude for scattering four *gauge bosons* contains only v and m_h : no trace of the gauge sector at all. That is already the equivalence theorem, staring at you from the formula.

Added explanation (not in the original notes) — Cross-check against the Goldstone calculation the notes refer to

The magenta remark points back to the $SO(2)$ Goldstone computation of an earlier lecture. It is worth reproducing in two lines, because the agreement is exact and not approximate. Before gauging, with $\Phi = \frac{1}{\sqrt{2}}(v + h)e^{i\pi/v}$,

$$|\partial_\mu \Phi|^2 = \frac{1}{2}(\partial_\mu h)(\partial^\mu h) + \frac{1}{2} \left(1 + \frac{h}{v}\right)^2 (\partial_\mu \pi)(\partial^\mu \pi) \supset \frac{h}{v} (\partial_\mu \pi)(\partial^\mu \pi),$$

so the $h\pi\pi$ vertex is $-is/v$ for two incoming Goldstones of invariant mass s , and the s -channel exchange gives $\left(-\frac{is}{v}\right)^2 \frac{i}{s - m_h^2} = -\frac{i}{v^2} \frac{s^2}{s - m_h^2}$. Summing s, t, u reproduces the boxed result *term by term*. Note that $SO(2) \cong U(1)$ — the notes use the two names for the same group interchangeably.

What the agreement means physically. Two calculations that look nothing alike — one with four massive spin-1 particles and their polarisation vectors, one with four massless scalars — give the identical function of s, t, u . The only way that can happen is if they are, at high energy, computations about the same degrees of freedom. This is the **Goldstone boson equivalence theorem**:

$$\mathcal{M}(A_L(k_1) \cdots) = \mathcal{M}(\pi(k_1) \cdots) + \mathcal{O}\left(\frac{m_A}{E}\right),$$

and it is the quantitative content of the phrase “inside the belly of the gauge boson” from [p. 6*****].

One consequence worth keeping. At high energy the *longitudinal* gauge bosons are the strongly-coupled part of the theory — they remember the scalar sector, not the gauge sector. Whenever one wants to probe the mechanism of symmetry breaking experimentally, the observable to look at is longitudinal vector boson scattering. That is not a slogan; it is this formula.

PART II — The non-abelian Higgs mechanism [p. 10*****]–[p. 13*****]

3 The construction, step by step

[p. 10*****]–[p. 11*****]

→ THE NON-ABELIAN HIGGS MECHANISM

We consider the case of a non-abelian $SU(N)$ symmetry.
The construction goes through the following points.

1) We start from the Lagrangian

$$\mathcal{L} = (\partial_\mu \Phi^\dagger)(\partial^\mu \Phi) - V(\Phi^\dagger \Phi)$$

$$V(\Phi^\dagger \Phi) = -\mu^2 \Phi^\dagger \Phi + \lambda (\Phi^\dagger \Phi)^2$$

Invariant under a **global** $SU(N)$ symmetry according to which Φ transforms as a **fundamental** object

$$\Phi(x) \longrightarrow \Phi'(x) = \exp(-i\alpha_A T^A) \Phi(x)$$

$\mu^2 > 0, \lambda > 0$: the potential is tailored to trigger SSB

2) We promote the symmetry to be a local gauge invariance. The theory is

$$\mathcal{L} = -\frac{1}{4} F_{\mu\nu}^A F^{A,\mu\nu} + (D_\mu \Phi)^\dagger (D^\mu \Phi) - \underbrace{V(\Phi^\dagger \Phi)}_{\text{same as before}}$$

$$D_\mu \Phi = \partial_\mu \Phi + ig G_\mu^A T^A \Phi$$

Added explanation (not in the original notes) — renumbering

The handwritten steps read “1)” and then “3)” on this sheet, with 3) and 4) following on [p. 11*****]. The second step is plainly “2)”; renumbered above so that the four steps run 1–2–3–4.

The theory is invariant under combined gauge transformations

$$\left\{ \begin{array}{l} \Phi(x) \longrightarrow \underbrace{\exp(-i\alpha_A(x) T^A)}_{\equiv U(x)} \Phi(x) \\ G_\mu^A T^A \longrightarrow U(x) G_\mu^A T^A U(x)^{-1} + \frac{i}{g} (\partial_\mu U) U^{-1} \end{array} \right.$$

3) The theory undergoes spontaneous breaking

There is a **manifold of vacua** described by

$$\Phi^\dagger \Phi = \frac{\mu^2}{2\lambda} \equiv \frac{v^2}{2}$$

All field configurations that minimise the energy satisfy this condition

↪ We choose a **vacuum representative** $\langle \Phi \rangle \equiv \Phi_0$

4) In principle, we should now introduce shifted fields and compute the Lagrangian in the broken phase

↪ For the argument we want to make, this is **not strictly necessary**. Let us just **evaluate the Lagrangian on the vacuum** Φ_0 .

On the vacuum, assuming only dynamical gauge fields, we have:

Added explanation (not in the original notes) — Why evaluating on the vacuum is legitimate — and what it does and does not give

This is a real shortcut and it is worth being clear about what it buys.

The logic. A mass term for a gauge field is, by definition, the piece of the Lagrangian quadratic in G_μ^A with no derivatives and no other fields. Such a term can only come from $|D_\mu \Phi|^2$ with *both* derivatives replaced by their gauge pieces and Φ replaced by its vev. Setting $\Phi \rightarrow \Phi_0$ (a constant) kills $\partial_\mu \Phi$ outright and projects onto exactly that piece. So the mass matrix is obtained *exactly*, with no approximation.

What is thrown away. Everything else: the Higgs kinetic term, the Higgs self-couplings, and every interaction between the physical scalars and the gauge fields. Those all require the shifted field $\Phi = \Phi_0 + \text{fluctuations}$. The two worked examples later in this scan *do* carry out the full shift, precisely so that those interactions appear — and they are needed for the $W_L W_L$ discussion on [p. 19*****]–[p. 21*****].

The one thing to be careful about. Evaluating on the vacuum does *not* tell you which linear combinations of G_μ^A are the mass eigenstates unless you then diagonalise $M_{\alpha\beta}^2$. In the two examples here the matrix comes out diagonal (or proportional to the identity) so the issue is invisible — but in the Standard Model it is not, and the W^3 – B mixing on [p. 6*****] is exactly this problem. See Section 18.

4 Evaluating on the vacuum: the Gram mass matrix

[p. 12*****]-[p. 13*****]

$$\mathcal{L} = -\frac{1}{4}F_{\mu\nu}^A F^{A,\mu\nu} + \underbrace{(D_\mu \Phi_0)^\dagger (D^\mu \Phi_0)}_{\text{we focus on this term}} - \underbrace{V_{\min}}_{\substack{\text{we can set} \\ V_{\min} \text{ to zero}}}$$

[p. 12*****]

$$D_\mu \Phi_0 = ig G_\mu^A T^A \Phi_0 ; \quad (D_\mu \Phi_0)^\dagger = -ig \Phi_0^\dagger T^A G_\mu^A$$

$$\rightsquigarrow (D_\mu \Phi_0)^\dagger (D^\mu \Phi_0) = g^2 G_\mu^A G^{B,\mu} \underbrace{(\Phi_0^\dagger T^A T^B \Phi_0)}_{\substack{\text{non-vanishing only in the} \\ \text{case of broken generators}}} = g^2 G_\mu^\alpha G^{\beta,\mu} (\Phi_0^\dagger X^\alpha X^\beta \Phi_0)$$

$$\rightsquigarrow (D_\mu \Phi_0)^\dagger (D^\mu \Phi_0) = g^2 G_\mu^\alpha G^{\beta,\mu} (\Phi_0^\dagger X^\alpha X^\beta \Phi_0)$$

We thus write

$$\mathcal{L} = -\frac{1}{4} \underbrace{F_{\mu\nu}^A F^{A,\mu\nu}}_{(1)} + \frac{1}{2} \underbrace{\left[2g^2 (\Phi_0^\dagger X^\alpha X^\beta \Phi_0) \right]}_{(2)} G_\mu^\alpha G^{\beta,\mu}$$

(1) Let us separate in the sum:

(2) This plays the role of a **mass term** (quadratic in the gauge fields, no derivatives)

$$-\frac{1}{4}F_{\mu\nu}^a F^{a,\mu\nu} - \frac{1}{4}F_{\mu\nu}^\alpha F^{\alpha,\mu\nu}$$

Added explanation (not in the original notes) — Why only the broken generators survive — the step the notes compress

The magenta remark on this sheet is the single most important line of the whole construction, and it is written very fast. Unpacked:

An *unbroken* generator is defined by $T^a \Phi_0 = 0$ — that is what “the vacuum is invariant under H ” means. So in the sum over A, B in $\Phi_0^\dagger T^A T^B \Phi_0$:

$$T^a \Phi_0 = 0 \implies \Phi_0^\dagger T^A T^a \Phi_0 = 0 \text{ and } \Phi_0^\dagger T^a T^B \Phi_0 = (T^a \Phi_0)^\dagger T^B \Phi_0 = 0$$

(using hermiticity of the generators). Every term with at least one unbroken index dies. Only α, β running over the *broken* generators X^α survives — which is exactly the replacement $T^A \rightarrow X^\alpha$ made in the third line.

The physical statement. A gauge boson gets a mass if and only if its generator fails to annihilate the vacuum. Gauge bosons of the unbroken subgroup stay massless, and they stay massless *exactly*, protected by the surviving gauge symmetry — not approximately, not up to corrections. In the Standard Model this is why the photon is massless to all orders.

Let us focus on the free-theory limit:

[p. 13*****]

$$\begin{aligned}\mathcal{L}_{\text{FREE}} = & \underbrace{-\frac{1}{4}(\partial_\mu G_\nu^a - \partial_\nu G_\mu^a)(\partial^\mu G^{a,\nu} - \partial^\nu G^{a,\mu})}_{(1)} \\ & \underbrace{-\frac{1}{4}(\partial_\mu G_\nu^\alpha - \partial_\nu G_\mu^\alpha)(\partial^\mu G^{\alpha,\nu} - \partial^\nu G^{\alpha,\mu})}_{(2)} \\ & \underbrace{+\frac{1}{2}(2g^2\Phi_0^\dagger X^\alpha X^\beta \Phi_0)G_\mu^\alpha G^{\beta,\mu}}_{(2)}\end{aligned}$$

(2) The remaining $\dim G - \dim H$ gauge bosons, those associated to broken directions, get a mass term.

(1) $\dim H$ gauge bosons, those associated to unbroken directions, remain massless.

The mass term is given by

$$M_{\alpha\beta}^2 = 2g^2 \Phi_0^\dagger X^\alpha X^\beta \Phi_0$$

→ This is a $(\dim G - \dim H)$ -dimensional (mass) matrix.

Mathematically, it is called a Gram matrix, since it is constructed by means of **inner products of vectors**.

It is hermitian and, since the vectors $X^\alpha \Phi_0$ are linearly independent, is positive definite (eigenvalues strictly positive).

Added explanation (not in the original notes) — The Gram matrix argument, spelled out — this is where the theorem is proved

The notes state three properties in one breath. Each one is doing work.

It is a Gram matrix. Define the vectors $w^\alpha \equiv X^\alpha \Phi_0$, one for each broken generator. Then

$$M_{\alpha\beta}^2 = 2g^2 (X^\alpha \Phi_0)^\dagger (X^\beta \Phi_0) = 2g^2 \langle w^\alpha, w^\beta \rangle,$$

using $(X^\alpha)^\dagger = X^\alpha$. A matrix of inner products of a set of vectors is precisely what “Gram matrix” means.

Hermitian. $(M_{\alpha\beta}^2)^* = \langle w^\beta, w^\alpha \rangle = M_{\beta\alpha}^2$. So it can be diagonalised with real eigenvalues and an orthonormal eigenbasis — i.e. the mass eigenstates are real linear combinations of the G_μ^α .

Positive definite — and why this is the whole point. The standard fact about Gram matrices: M^2 is positive semi-definite always, since for any vector c_α ,

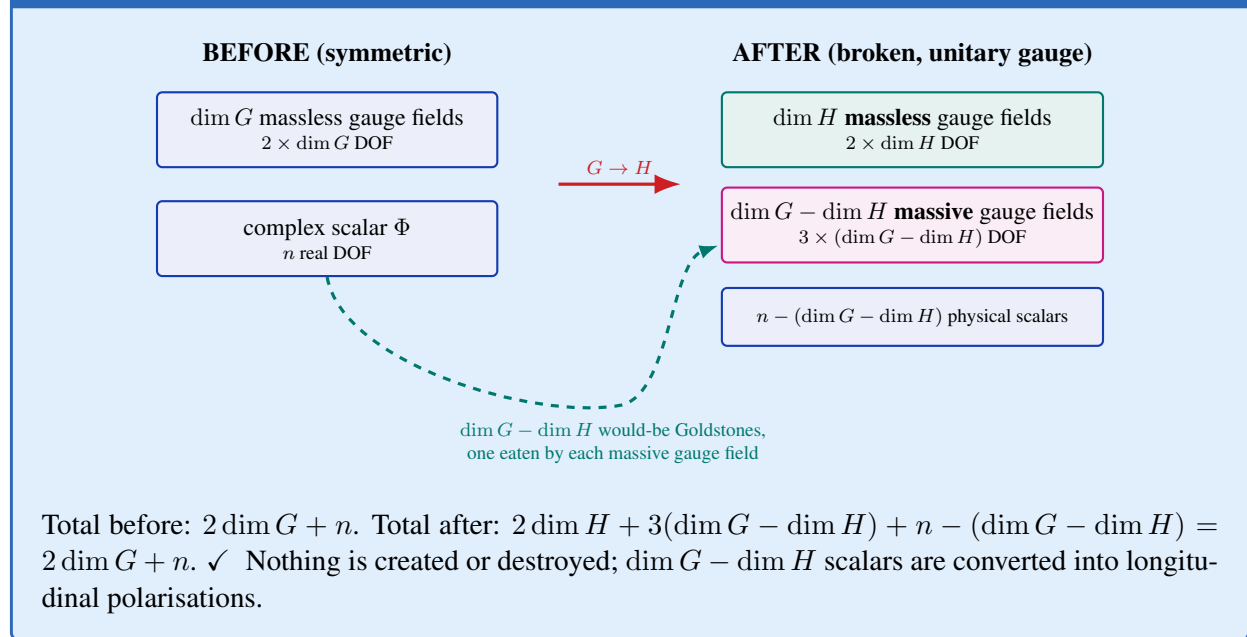
$$c_\alpha^* M_{\alpha\beta}^2 c_\beta = 2g^2 \left\| \sum_\alpha c_\alpha w^\alpha \right\|^2 \geq 0,$$

and it is *strictly* positive precisely when the w^α are linearly independent (so that no non-trivial combination has zero norm). And they *are* linearly independent: a relation $\sum_\alpha c_\alpha X^\alpha \Phi_0 = 0$ would say that the generator $\sum_\alpha c_\alpha X^\alpha$ annihilates the vacuum, i.e. that it is unbroken — but the X^α were defined as a basis of the broken directions, complementary to the unbroken ones. Contradiction.

So the theorem is: every broken direction gets a *strictly positive* mass-squared. Not one of them slips through massless. Combined with the previous box (unbroken $\Rightarrow$ exactly massless), the count $\dim G - \dim H$ is exact and saturated.

Bookkeeping check. $\dim G - \dim H$ is also the number of Goldstone bosons the breaking would have produced had the symmetry been global (Lecture 27). One eaten Goldstone per massive gauge boson: the accounting is one-to-one, as it must be.

Added explanation (not in the original notes) — Figure — the degree-of-freedom ledger in the general case



We thus find that

$\dim G - \dim H$ gauge bosons become massive

[p. 14*****]

We discuss **two examples**.

PART III — Example 1: $SU(2) \rightarrow$ nothing [p. 14***]–[p. 21*****]**

5 Setup and breaking pattern

[p. 14*****]–[p. 15*****]

$$SU(2) \longrightarrow \text{NOTHING}$$

The Lagrangian is

$$\mathcal{L} = -\frac{1}{4}W_{\mu\nu}^A W^{A,\mu\nu} + (D_\mu H)^\dagger (D^\mu H) + \underbrace{\mu^2 H^\dagger H - \lambda (H^\dagger H)^2}_{\equiv -V(H^\dagger H)}$$

$$W_{\mu\nu}^A = \partial_\mu W_\nu^A - \partial_\nu W_\mu^A - g \varepsilon_{ABC} W_\mu^B W_\nu^C$$

H is a **complex doublet**, and it transforms according to the **fundamental of $SU(2)$**

The symmetry is spontaneously broken since we have a degeneracy of vacua described by

$$H^\dagger H = \frac{\mu^2}{2\lambda} = \frac{v^2}{2}, \quad v^2 \equiv \frac{\mu^2}{\lambda}$$

We consider the vacuum choice

$$\langle H \rangle \equiv H_0 = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ v \end{pmatrix}$$

There is **no subgroup** of G that leaves this vacuum invariant

If we consider the generators $\frac{\sigma^A}{2} \equiv T^A$, we find

$$\frac{\sigma^A}{2} \begin{pmatrix} 0 \\ v \end{pmatrix} \neq 0, \quad A = 1, 2, 3 \quad \blacktriangleright \quad \text{all generators are broken}$$

The spontaneous breaking delivers 3 Goldstone bosons.

We introduce the parametrisation

$$H(x) = \underbrace{\exp\left(\frac{i\pi^a(x)\sigma^a}{2v}\right)}_{\text{Goldstone matrix}} \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ v + h(x) \end{pmatrix}$$

The theory is also invariant under combined gauge transformations

$$H(x) \longrightarrow H'(x) = \exp\left(-i\alpha_A(x)\frac{\sigma^A}{2}\right)H(x) \equiv U(x)H(x)$$

$$W_\mu^A \frac{\sigma^A}{2} \longrightarrow W_\mu'^A \frac{\sigma^A}{2} = U W_\mu^A \frac{\sigma^A}{2} U^{-1} + \frac{i}{g}(\partial_\mu U)U^{-1}$$

Added explanation (not in the original notes) — Checking “all generators are broken” explicitly, and the group theory behind it

The three checks. With $H_0 = \frac{1}{\sqrt{2}}(0, v)^T$:

$$\frac{\sigma^1}{2}H_0 = \frac{1}{2\sqrt{2}} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 0 \\ v \end{pmatrix} = \frac{1}{2\sqrt{2}} \begin{pmatrix} v \\ 0 \end{pmatrix} \neq 0, \quad \frac{\sigma^2}{2}H_0 = \frac{1}{2\sqrt{2}} \begin{pmatrix} -iv \\ 0 \end{pmatrix} \neq 0,$$

$$\frac{\sigma^3}{2}H_0 = \frac{1}{2\sqrt{2}} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} 0 \\ v \end{pmatrix} = \frac{1}{2\sqrt{2}} \begin{pmatrix} 0 \\ -v \end{pmatrix} \neq 0.$$

All three fail to annihilate H_0 : **all broken**. Note in particular that σ^3 , which *does* preserve the vacuum *direction* up to a phase, still does not annihilate it — and annihilation is the criterion. (Contrast the Standard Model on [p. 4*****], where a *combination* $T^3 + Y$ does annihilate.)

The geometric statement. H is a complex doublet, i.e. four real fields, and the vacuum condition $H^\dagger H = v^2/2$ carves out a three-sphere S^3 in $\mathbb{R}^4$. Now $SU(2) \cong S^3$ as a manifold, and $SU(2)$ acts transitively on the sphere of fixed radius with no fixed points. So the vacuum manifold *is* the whole group, $\mathcal{M}_0 \cong SU(2)/1 \cong S^3$, and $\dim G - \dim H = 3 - 0 = 3$ Goldstones. ✓

The custodial bonus (why the three W ’s will come out degenerate). The Lagrangian depends on H only through $H^\dagger H$, which is invariant under the *full* $SO(4)$ acting on the four real components — and $SO(4) \cong SU(2)_L \times SU(2)_R$. The vev breaks this to the diagonal $SU(2)_{\text{custodial}}$. The three gauge bosons sit in a triplet of that surviving *global* symmetry, so they are forced to be degenerate. That is why $m_{W^1} = m_{W^2} = m_{W^3}$ on [p. 19*****] without any calculation — and, in the Standard Model, it is why $\rho = m_W^2/(m_Z^2 \cos^2 \theta_W) = 1$ (Section 18).

6 The unitary gauge

[p. 16*****]

We consider again a specific choice of gauge such that

$$\alpha_A(x) = \frac{\pi^A(x)}{v} \quad \text{unitary gauge}$$

[p. 16*****]

We have in unitary gauge

$$H_u = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ v + h(x) \end{pmatrix}$$

The Goldstone bosons **disappear** from H in unitary gauge

$$W_u^\mu \equiv U W^\mu U^{-1} + \frac{i}{g} (\partial^\mu U) U^{-1}, \quad U = \exp\left(\frac{i\pi^A \sigma^A}{2v}\right)$$

$$W_u^\mu = W_{u,\mu}^{A,\mu} \frac{\sigma^A}{2}$$

$$W^\mu \equiv W^{A,\mu} \frac{\sigma^A}{2}$$

By definition and construction, the Lagrangian takes the **same form** when written in terms of fields in the unitary gauge

$$\mathcal{L} = -\frac{1}{4} W_{u,\mu\nu}^A W_u^{A,\mu\nu} + (D_\mu H_u)^\dagger (D^\mu H_u) - V(H_u^\dagger H_u)$$

with $H_u = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ v + h \end{pmatrix}$

↪ From now on, we omit the subscript “ u ”

Added explanation (not in the original notes) — Three Goldstones, three gauge parameters — the counting that makes this work

The abelian case needed one arbitrary function $\alpha(x)$ to remove one Goldstone $\pi(x)$. Here there are three Goldstones $\pi^{1,2,3}(x)$ and, conveniently, exactly three arbitrary gauge functions $\alpha^{1,2,3}(x)$. The match is not an accident: there is one gauge parameter per generator of G , and one would-be Goldstone per *broken* generator, so the gauge freedom is always at least enough. When H is a proper subgroup the leftover $\dim H$ parameters remain as an honest, unbroken gauge symmetry (Part IV).

The non-abelian subtlety the notes glide over. In the abelian case the cancellation $e^{-i\alpha} e^{i\pi/v} = 1$ was exact because phases commute. Here $U^{-1}(\alpha) \exp(i\pi^a \sigma^a / 2v) = 1$ requires $\alpha_A(x) = \pi^A(x)/v$ only because both matrices are exponentials of the *same* linear combination — one is literally the inverse of the other. At higher orders in the fields the relation between α_A and π^A is deformed by Baker–Campbell–Hausdorff terms, but that is a field redefinition of the π ’s, and by the equivalence theorem of Lecture 25 it does not affect the S-matrix. The gauge *is* reachable; only the explicit formula for $\alpha_A(\pi)$ is more complicated than stated.

Why the Lagrangian “takes the same form”. Because the map $(H, W) \rightarrow (H_u, W_u)$ is a gauge transformation, and the Lagrangian was *built* to be gauge invariant. Nothing is being assumed — this is what gauge invariance is for.

7 The broken phase: potential and kinetic terms

[p. 17*****]–[p. 18*****]

POTENTIAL

This is the usual computation. We find

$$V(h) = \frac{1}{2}(2\mu^2)h^2 + \lambda v h^3 + \frac{\lambda}{4}h^4 - \frac{\mu^4}{4\lambda}$$

$$m_h^2 = 2\mu^2 = 2\lambda v^2 \quad \text{Higgs mass squared}$$

KINETIC TERM

Let us elaborate $(D_\mu H)^\dagger (D^\mu H)$

$$\partial_\mu H^\dagger - ig H^\dagger \frac{\sigma^A}{2} W_\mu^A \qquad \partial^\mu H + ig W^{A,\mu} \frac{\sigma^A}{2} H$$

$$\begin{aligned} (D_\mu H)^\dagger (D^\mu H) &= \left(\partial_\mu H^\dagger - ig W_\mu^A H^\dagger \frac{\sigma^A}{2} \right) \left(\partial^\mu H + ig W^{A,\mu} \frac{\sigma^A}{2} H \right) = \\ &= (\partial_\mu H^\dagger) (\partial^\mu H) + \frac{g^2}{4} W_\mu^A W^{B,\mu} H^\dagger \sigma^A \sigma^B H \\ &\quad - \underbrace{\frac{ig}{2} W_\mu^A H^\dagger \sigma^A (\partial^\mu H) + \frac{ig}{2} (\partial_\mu H^\dagger) W^{A,\mu} \sigma^A H}_{\text{let us focus on these two terms}} \quad (\otimes) \end{aligned}$$

$$\frac{ig}{2} W_\mu^A \left[(\partial^\mu H^\dagger) \sigma^A H - H^\dagger \sigma^A (\partial^\mu H) \right] =$$

$$= \frac{ig}{2} W_\mu^A \left[(0, \partial^\mu h) \frac{\sigma^A}{2} \begin{pmatrix} 0 \\ v+h \end{pmatrix} - (0, v+h) \frac{\sigma^A}{2} \begin{pmatrix} 0 \\ \partial^\mu h \end{pmatrix} \right] =$$

$$= \frac{ig}{2} W_\mu^3 \left[-\frac{1}{2}(v+h)(\partial^\mu h) + (v+h)\frac{1}{2}(\partial^\mu h) \right] = 0$$

Added explanation (not in the original notes) — reading the two lines above

Two things are worth pinning down, because the factors of $\frac{1}{2}$ drift in the original.

Why $A = 1, 2$ drop out. σ^1 and σ^2 are purely off-diagonal, so acting on $(0, v+h)^T$ they produce a vector with only an *upper* component. Contracting with the row vector $(0, \partial^\mu h)$, which has only a lower entry, gives zero. Only the diagonal σ^3 can connect them.

Why $A = 3$ also drops out. With $\sigma^3 = \text{diag}(1, -1)$ the two surviving terms are $-\frac{1}{2}(v+h)(\partial^\mu h)$ and $+\frac{1}{2}(v+h)(\partial^\mu h)$ — equal in magnitude, opposite in sign, because the bracket is an *antisymmetrised* combination $X^\dagger Y - Y^\dagger X$ of two real quantities. It vanishes identically.

What this means. There is **no $W_\mu \partial^\mu h$ kinetic mixing** in the broken phase. This is the non-abelian version of the same statement made in Lecture 29 [p. 4*****]: in unitary gauge the spectrum can be read straight off the Lagrangian without diagonalising away a gauge–scalar mixing first. (In an

R_ξ gauge the analogous term $\propto g v W_\mu^A \partial^\mu \pi^A$ is present and is cancelled by the gauge-fixing term.)

$$\begin{aligned}
 (\otimes) &= (\partial_\mu H^\dagger)(\partial^\mu H) + \frac{g^2}{4} W_\mu^A W^{B,\mu} H^\dagger \sigma^A \sigma^B H = \\
 &= \frac{1}{2} (\partial_\mu h)(\partial^\mu h) + \frac{g^2}{4} W_\mu^A W^{B,\mu} H^\dagger \left(\delta_{AB} \mathbf{1}_{2 \times 2} + \underbrace{i \varepsilon_{ABC} \sigma^C}_{(\dagger)} \right) H = \\
 &= \frac{1}{2} (\partial_\mu h)(\partial^\mu h) + \frac{g^2}{4} W_\mu^A W^{B,\mu} \delta_{AB} \underbrace{H^\dagger H}_{(\dagger)} = \\
 &= \frac{1}{2} (\partial_\mu h)(\partial^\mu h) + \frac{g^2 v^2}{8} W_\mu^A W^{A,\mu} + \frac{g^2}{8} W_\mu^A W^{A,\mu} (h^2 + 2 h v)
 \end{aligned}$$

Vanishes because ε_{ABC} is antisymmetric in $A \leftrightarrow B$ while $g^{\mu\nu} W_\mu^A W_\nu^B$ is symmetric
 $(\dagger) \quad H^\dagger H = \frac{1}{2} \begin{pmatrix} 0 & v+h \\ v+h & 0 \end{pmatrix} = \frac{1}{2} \begin{pmatrix} v+h & 0 \\ 0 & v+h \end{pmatrix}$
 $h^2 = \frac{1}{2} (v^2 + h^2 + 2vh)$

We thus arrive at the following Lagrangian in the broken phase

Added explanation (not in the original notes) — The identity $\sigma^A \sigma^B = \delta_{AB} \mathbf{1} + i \varepsilon_{ABC} \sigma^C$, and why the second piece must die

The Pauli identity is the engine of this whole page. Splitting the product into its symmetric and antisymmetric parts,

$$\sigma^A \sigma^B = \underbrace{\frac{1}{2} \{\sigma^A, \sigma^B\}}_{=\delta_{AB} \mathbf{1}} + \underbrace{\frac{1}{2} [\sigma^A, \sigma^B]}_{=i \varepsilon_{ABC} \sigma^C},$$

and the object it is contracted against, $W_\mu^A W^{B,\mu}$, is manifestly *symmetric* under $A \leftrightarrow B$ (it is a product of two commuting classical fields with the Lorentz indices already contracted). A symmetric tensor contracted with an antisymmetric one gives zero. Only δ_{AB} survives — which is exactly why the mass matrix comes out **proportional to the identity**, i.e. why all three W 's are degenerate.

Cross-check against the general formula of [p. 13***].** There, $M_{\alpha\beta}^2 = 2g^2 \Phi_0^\dagger X^\alpha X^\beta \Phi_0$. Here $X^A = \sigma^A/2$ and $\Phi_0 = H_0 = \frac{1}{\sqrt{2}}(0, v)^T$, so

$$M_{AB}^2 = 2g^2 \cdot \frac{1}{4} H_0^\dagger \sigma^A \sigma^B H_0 = \frac{g^2}{2} \delta_{AB} H_0^\dagger H_0 = \frac{g^2}{2} \delta_{AB} \frac{v^2}{2} = \frac{g^2 v^2}{4} \delta_{AB}.$$

Identical to what [p. 19*****] will box as $m_W^2 = \frac{1}{4} g^2 v^2$. ✓ The Gram matrix is $\frac{g^2 v^2}{4} \mathbf{1}_{3 \times 3}$: hermitian ✓, positive definite ✓, three strictly positive eigenvalues ✓ — all as the theorem promised.

8 The broken-phase Lagrangian, and $W_L W_L$ scattering

[p. 19*****]–[p. 21*****]

$$\begin{aligned}\mathcal{L} = & \frac{1}{2}(\partial_\mu h)(\partial^\mu h) \underbrace{-\frac{1}{2}(2\mu^2)h^2 - \lambda v h^3 - \frac{\lambda}{4}h^4}_{(1)} \\ & - \frac{1}{4}W_{\mu\nu}^A W^{A,\mu\nu} + \underbrace{\frac{1}{2}\left(\frac{g^2 v^2}{4}\right)W_\mu^A W^{A,\mu}}_{(2)} \\ & + \underbrace{\frac{1}{2}\left(\frac{g^2}{4}\right)(h^2 + 2hv)W_\mu^A W^{A,\mu}}_{(3)}\end{aligned}$$

[p. 19*****]

(1) ▲ The **Higgs boson** is massive, with mass $m_h^2 = 2\mu^2$

(2) The **three gauge bosons** are massive, with **common mass**

(3) There are **interactions** between the Higgs and the gauge bosons

$$m_W^2 = \frac{1}{4}g^2 v^2$$

↪ The **3 Goldstone fields** are “eaten” by the gauge fields that are now massive

COMMENT

Since the gauge fields are now massive, we can consider a **longitudinally polarised scattering**

$$W_L(p_1) + W_L(p_2) \longrightarrow W_L(p_3) + W_L(p_4)$$

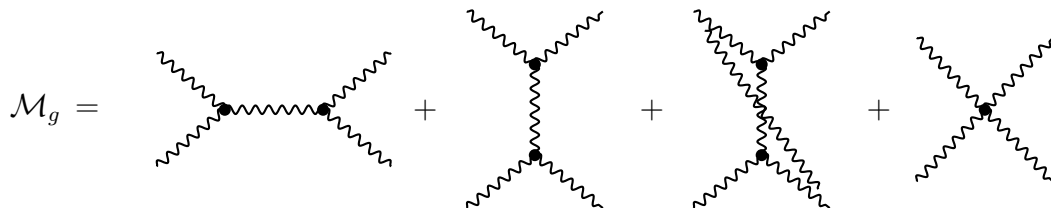
Let us discuss various contributions (at the tree level)

1) First, we consider the Lagrangian

$$\mathcal{L} \supset -\frac{1}{4}W_{\mu\nu}^A W^{A,\mu\nu} + \frac{1}{2}m_W^2 W_\mu^A W^{A,\mu}$$

[p. 20*****]

These are “pure gauge” contributions, that give rise to the diagrams



$$\simeq \mathcal{O}\left(\frac{g^2 E^2}{m_W^2}\right) \text{ in the UR limit}$$

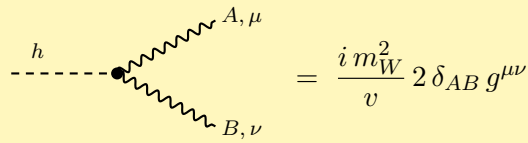
$$\rightsquigarrow \text{Violation of unitarity at } E = \mathcal{O}\left(\frac{m_W}{g}\right)$$

This is the same violation of unitarity that we have discussed in **pure Yang–Mills + mass term**

2) We now have an **additional interaction**

$$\mathcal{L} \supset \frac{g^2}{8}(h^2 + 2hv)g^{\mu\nu}W_\mu^A W_\nu^A$$

The cubic vertex contributes to the process



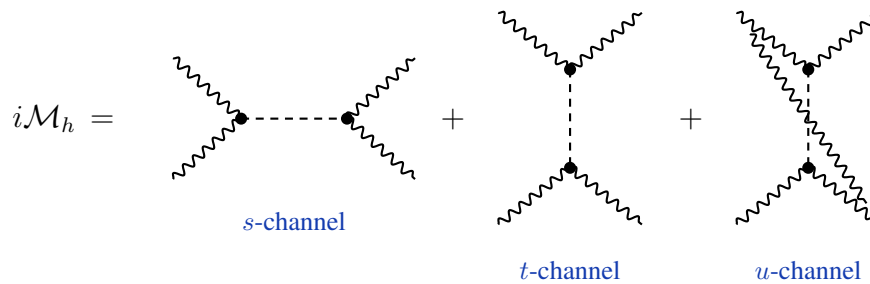
$$= \frac{i m_W^2}{v} 2 \delta_{AB} g^{\mu\nu}$$

Added explanation (not in the original notes) — checking the vertex factor

From $\mathcal{L} \supset \frac{g^2}{8} \cdot 2 h v W_\mu^A W^{A,\mu} = \frac{g^2 v}{4} h W_\mu^A W^{A,\mu}$, the Feynman rule is $i \times \frac{g^2 v}{4} \times 2 \times \delta_{AB} g^{\mu\nu} = \frac{i g^2 v}{2} \delta_{AB} g^{\mu\nu}$ (the 2 from the two ways of assigning the gauge fields to the external legs). Since $m_W^2 = \frac{g^2 v^2}{4}$, we have $\frac{g^2 v}{2} = \frac{2 m_W^2}{v}$, so the vertex is $\frac{2 i m_W^2}{v} \delta_{AB} g^{\mu\nu}$ — exactly as boxed. ✓

Note the form of the answer: the coupling of the Higgs to a pair of gauge bosons is $2 m_W^2 / v$. *The strength of the hWW coupling is fixed by the gauge boson mass.* This is the sharpest prediction of the mechanism, and it is what is measured at the LHC.

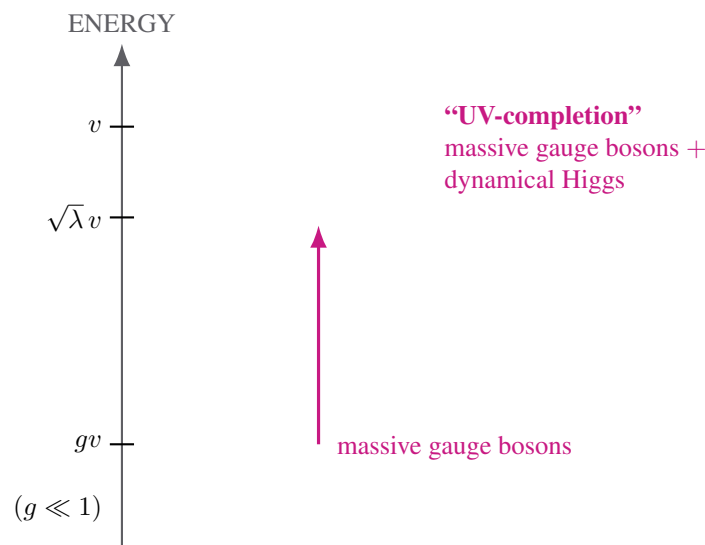
We have the additional diagrams



[p. 21]*****

The final amplitude is $i(\mathcal{M}_h + \mathcal{M}_g)$ and it is possible to show that, thanks to the new diagrams, it does not grow with energy

↪ In the jargon, we say that “the Higgs unitarises the longitudinal WW scattering”



Added explanation (not in the original notes) — What “it does not grow with energy” means, in numbers

The notes say “it is possible to show”. It is, and the structure is worth seeing because it is the whole reason the Higgs was expected.

Step 1 — what the gauge diagrams alone do. Each external longitudinal leg carries $\varepsilon_L \simeq p/m_W$, so four of them supply E^4/m_W^4 . The triple and quartic gauge vertices supply momenta and g ’s. The leading behaviour of the sum of the four gauge diagrams is

$$\mathcal{M}_g \sim \underbrace{\mathcal{O}\left(\frac{g^2 E^4}{m_W^4}\right)}_{\text{cancels among the four}} + \mathcal{O}\left(\frac{g^2 E^2}{m_W^2}\right).$$

The E^4 pieces cancel *among the gauge diagrams themselves* — a consequence of the Yang–Mills structure, i.e. of the Jacobi identity for ε_{ABC} . What survives is the E^2 growth quoted on [p. 20*****]. Setting $g^2 E^2/m_W^2 \sim 16\pi$ gives $E \sim \mathcal{O}(m_W/g) = \mathcal{O}(v)$: unitarity fails at the scale v , not at m_W .

Step 2 — what the Higgs diagrams do. The three h -exchange diagrams each grow like E^2/v^2 as well, with the coefficient fixed by the vertex $2m_W^2/v$ — and, using $m_W = gv/2$, $\frac{(2m_W^2/v)^2}{m_W^4} = \frac{4}{v^2}$,

the *same* size as the leftover gauge growth. They cancel it. The residual amplitude is a constant, of order m_h^2/v^2 and m_W^2/v^2 .

Step 3 — the equivalence-theorem shortcut. One does not have to grind through the polarisation vectors at all. By [p. 9^{*****}], at $E \gg m_W$ the amplitude for four W_L 's equals the amplitude for four Goldstones — and *that* calculation has no gauge diagrams, only Higgs exchange, and manifestly tends to a constant because $s + t + u$ is fixed. Same answer, three lines instead of thirty.

Why the numerical coincidence is not a coincidence. The cancellation requires the hWW coupling to be exactly $2m_W^2/v$ and the W mass to be exactly $gv/2$ — both from the *same* term $|D_\mu H|^2$ with the *same* g and the *same* v . Rescale the hWW coupling by even a few percent and the cancellation is spoiled, and $W_L W_L$ scattering grows again until some new physics intervenes. This is exactly what LHC coupling measurements are testing.

Added explanation (not in the original notes) — Reading the energy ladder — three scales, not two

The sketch on [p. 21^{*****}] packs the whole story into one axis. With $g \ll 1$ and λ generic:

$$\underbrace{m_W = \frac{1}{2}gv}_{\text{lightest}} \ll \underbrace{m_h = \sqrt{2\lambda}v}_{\text{intermediate}} \lesssim \underbrace{v}_{\text{the scale of SSB}}$$

- **Below m_W :** nothing to see; the massive vectors are not produced.
- **Between m_W and m_h :** the effective theory is “massive gauge bosons and nothing else” — i.e. Proca fields, or equivalently a gauged non-linear sigma model. It is non-renormalisable and its $W_L W_L$ amplitude grows like $g^2 E^2 / m_W^2$. This is the magenta arrow.
- **At m_h and above:** the Higgs becomes a propagating particle, its exchange diagrams switch on, and the growth is cut off. The theory is renormalisable and valid to arbitrarily high energy. This is the “UV-completion” label.

So the Higgs boson is, in the precise technical sense, the *UV completion* of the theory of massive vector bosons. The phrase in the notes is not loose language — the Higgs is what makes the low-energy effective theory extendable.

And if there were no Higgs? Then the growth continues until $E \sim 4\pi v$ and something else must appear: a strongly-coupled sector, resonances, technicolour. The LHC excluded that possibility by finding the scalar at 125 GeV.

PART IV — Example 2: the Georgi–Glashow model
 $SU(2) \rightarrow U(1)$ [p. 22*****]–[p. 29*****]

9 The adjoint scalar, and the invariant $\Phi^T \Phi$

[p. 22*****]

$$SU(2) \rightarrow U(1)$$

“GEORGI–GLASHOW MODEL”

[p. 22*****]

Consider a scalar field Φ that transforms according to the **adjoint** of $SU(2)$

$\hookrightarrow$ It means that Φ is a **three-dimensional vector of real fields**

$$\Phi(x) = \begin{pmatrix} \Phi^1(x) \\ \Phi^2(x) \\ \Phi^3(x) \end{pmatrix}$$

that transforms under the action of $SU(2)$ according to

$$\begin{aligned} \Phi^A(x) &\longrightarrow \Phi'^A(x) = \left(\exp(-i\alpha_B t^B) \right)_{AC} \Phi^C(x) \\ (t^A)_{BC} &= -i\varepsilon_{ABC} \end{aligned}$$

$\Phi^T \Phi$ is an invariant object

$\hookrightarrow$ Since $\Phi \rightarrow e^{-i\alpha_A t^A} \Phi$, we have

$$\Phi^T \longrightarrow \Phi^T (e^{-i\alpha_A t^A})^T = \Phi^T e^{-i\alpha_A (t^A)^T} = \Phi^T e^{+i\alpha_A t^A}, \quad \text{since } (t^A)^T = -t^A$$

We thus have

$$\Phi^T \Phi \longrightarrow \Phi^T \underbrace{e^{i\alpha_A t^A} e^{-i\alpha_A t^A}}_{=1} \Phi = \Phi^T \Phi$$

[p. 23*****]

Added explanation (not in the original notes) — Why the adjoint is a real vector, and why $(t^A)^T = -t^A$

The representation. The adjoint of $SU(2)$ acts on the Lie algebra itself, which is three-dimensional. In the basis where $(t^A)_{BC} = -i\varepsilon_{ABC}$ these are precisely the generators of rotations in $\mathbb{R}^3$ — the adjoint of $SU(2)$ is the vector representation of $SO(3)$. That is why Φ has three *real* components rather than two complex ones: a triplet, not a doublet.

The transpose property. ε_{ABC} is totally antisymmetric, so $(t^A)_{CB} = -i\varepsilon_{ACB} = +i\varepsilon_{ABC} = -(t^A)_{BC}$, i.e. $(t^A)^T = -t^A$. The generators are purely imaginary and antisymmetric — which is the same as saying $e^{-i\alpha t}$ is a *real orthogonal* matrix. Hence Φ stays real under the transformation, and $\Phi^T \Phi$ (rather than $\Phi^\dagger \Phi$) is the invariant. Everything in this section is $SO(3)$ geometry in disguise.

Contrast with Example 1. There, H was a *fundamental* (doublet, complex), the vacuum manifold was S^3 , and the group acted with no fixed points — so nothing survived. Here Φ is an *adjoint* (triplet, real), the vacuum manifold is a two-sphere S^2 in field space, and a rotation about the axis picked out by Φ_0 obviously leaves it alone — so a $U(1)$ survives. **The representation of the scalar determines the unbroken subgroup.** That is the single lesson of comparing the two examples.

10 Breaking pattern and gauging

[p. 23*****]-[p. 24*****]

We thus consider the theory based on the Lagrangian

$$\mathcal{L} = \frac{1}{2}(\partial_\mu \Phi)^T (\partial^\mu \Phi) - V(\Phi^T \Phi)$$

$$V(\Phi^T \Phi) = -\frac{\mu^2}{2} \Phi^T \Phi + \frac{\lambda}{4} (\Phi^T \Phi)^2$$

We have SSB since $\Phi^T \Phi = \frac{\mu^2}{\lambda} = v^2$ minimises the potential. We choose the vacuum

$$\langle \Phi \rangle = \Phi_0 = \begin{pmatrix} 0 \\ 0 \\ v \end{pmatrix}$$

Let us now check the **symmetry breaking pattern**

The generators are

$$t^1 = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & -i \\ 0 & i & 0 \end{pmatrix}, \quad t^2 = \begin{pmatrix} 0 & 0 & i \\ 0 & 0 & 0 \\ -i & 0 & 0 \end{pmatrix}, \quad t^3 = \begin{pmatrix} 0 & -i & 0 \\ i & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

and we find

$$t^1 \Phi_0 = \begin{pmatrix} 0 \\ -iv \\ 0 \end{pmatrix} \neq 0, \quad t^2 \Phi_0 = \begin{pmatrix} iv \\ 0 \\ 0 \end{pmatrix} \neq 0, \quad t^3 \Phi_0 = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} = 0$$

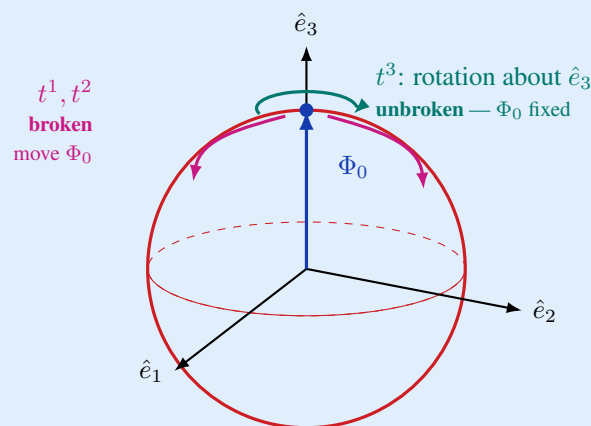
BROKEN

BROKEN

UNBROKEN

↪ **Rotations around the third axis (in field space) are unbroken**

Added explanation (not in the original notes) — Figure — the vacuum manifold is a sphere, and one rotation survives



$\mathcal{M}_0 \cong S^2$, radius v

The vacuum condition $\Phi^T \Phi = v^2$ is a sphere of radius v in the three-dimensional field space. Picking Φ_0 along $\hat{e}_3$ leaves exactly one rotation — the one about that very axis — unbroken. The other two

tilt the vector off the pole. This is the same picture as the $SO(3) \rightarrow SO(2)$ example of Lecture 28, now with the symmetry *gauged*.

Counting: $\dim G - \dim H = 3 - 1 = 2$. Two would-be Goldstones (the two directions you can move on S^2), so two gauge bosons become massive and one stays massless. $\mathcal{M}_0 \cong SU(2)/U(1) \cong S^2$. ✓

We thus have

$$SU(2) \longrightarrow SO(2) \cong U(1)$$

$\hookrightarrow$ This breaking delivers **2 Goldstone bosons**

[p. 24*****]

We now **gauge** the $SU(2)$ symmetry. We consider the Lagrangian

$$\mathcal{L} = -\frac{1}{4}W_{\mu\nu}^A W^{A,\mu\nu} + \frac{1}{2}(D_\mu\Phi)^T(D^\mu\Phi) - V(\Phi^T\Phi)$$

The covariant derivative is $D_\mu\Phi = \partial_\mu\Phi + ig W_\mu^A t^A\Phi$

The Lagrangian is, by construction, invariant under combined gauge transformation

$$\begin{cases} \Phi(x) \longrightarrow U(x)\Phi(x) \\ W_\mu \longrightarrow U W_\mu U^{-1} + \frac{i}{g}(\partial_\mu U)U^{-1} \end{cases} \quad U(x) = \exp(-i\alpha_A(x)t^A)$$

We introduce the parametrisation

$$\Phi(x) = \exp\left(\frac{i\pi^1 t^1 + i\pi^2 t^2}{v}\right) \begin{pmatrix} 0 \\ 0 \\ v + h(x) \end{pmatrix}$$

and eliminate the Goldstone fields via a **unitary gauge fixing along the broken directions**

[p. 25*****]

$$\Phi(x) = \begin{pmatrix} 0 \\ 0 \\ v + h(x) \end{pmatrix} \quad \text{unitary gauge}$$

Added explanation (not in the original notes) — Note the exponent: only the *broken* generators appear

The parametrisation uses $\exp(i(\pi^1 t^1 + \pi^2 t^2)/v)$ — t^3 is deliberately absent. This is the CCWZ lesson of Lecture 28 applied here: the Goldstone matrix is built from broken generators only, because the unbroken ones annihilate Φ_0 and would be pure redundancy ($e^{i\theta t^3}\Phi_0 = \Phi_0$).

Correspondingly, the gauge fixing uses only α_1, α_2 ; the third gauge parameter $\alpha_3(x)$ is *not* spent. It survives as a genuine, unbroken $U(1)$ gauge symmetry — which is why W_μ^3 remains massless and why the rest of this section is about identifying what that $U(1)$ does. Contrast Example 1, where all three parameters were used up and nothing survived.

11 The kinetic term and the mass spectrum

[p. 25*****]–[p. 26*****]

What happens in the broken phase?

The potential generates the “usual” massive Higgs mode. Let us investigate the **kinetic term**

$$\begin{aligned}
\frac{1}{2}(D_\mu\Phi)^T(D^\mu\Phi) &= \frac{1}{2}\left[\partial_\mu\Phi^T + ig W_\mu^A\Phi^T(t^A)^T\right]\left[\partial^\mu\Phi + ig W^{B,\mu}t^B\Phi\right] = \\
&= \frac{1}{2}\left[(\partial_\mu\Phi^T)(\partial^\mu\Phi) + ig W_\mu^A(\partial^\mu\Phi^T)t^A\Phi + ig W_\mu^A\Phi^T(t^A)^T(\partial^\mu\Phi) \right. \\
&\quad \left. - g^2 W_\mu^A W^{B,\mu}\Phi^T(t^A)^T t^B\Phi\right] = \quad \text{since } (t^A)^T = -t^A \\
&= \frac{1}{2}(\partial_\mu\Phi^T)(\partial^\mu\Phi) + \frac{ig}{2}W_\mu^A \underbrace{\left[(\partial^\mu\Phi^T)t^A\Phi - \Phi^T t^A(\partial^\mu\Phi)\right]}_{(\otimes)} \\
&\quad + \frac{g^2}{2}W_\mu^A W^{B,\mu} \underbrace{\Phi^T t^A t^B \Phi}_{(\otimes\otimes)} \quad \star =
\end{aligned}$$

Let us first compute $(\otimes)$

All these terms are $= 0$ (for $A = 1, 2, 3$) since, in components, we have for instance

$$\Phi^T t^A (\partial^\mu\Phi) = \Phi^B (t^A)_{BC} (\partial^\mu\Phi^C) = (v+h) \underbrace{(t^A)_{33}}_{=0} (\partial^\mu h) = 0$$

$$\text{since } (t^A)_{33} = -i\epsilon_{A33} = 0$$

We now compute $(\otimes \otimes)$

$$g^2 W_\mu^A W^{B,\mu} \underbrace{\Phi^T t^A t^B \Phi}_{\text{in components, this object is}}$$

[p. 26*****]

$$\begin{aligned} \Phi^a (t^A)_{ab} (t^B)_{bc} \Phi^c &= (-1) \Phi^a \varepsilon_{Aab} \varepsilon_{Bbc} \Phi^c = \\ &= \Phi^a (\delta_{AB} \delta_{ac} - \delta_{Ac} \delta_{aB}) \Phi^c = \\ &= \delta_{AB} \Phi^a \Phi^a - \Phi^B \Phi^A = \\ &= \delta_{AB} (v+h)^2 - \delta_{A3} \delta_{B3} (v+h)^2 \end{aligned}$$

$\hookrightarrow$ As a matrix, we thus find

$$\Phi^T t^A t^B \Phi = \begin{pmatrix} (v+h)^2 & 0 & 0 \\ 0 & (v+h)^2 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

$$= g^2 (v+h)^2 (W_\mu^1 W^{1,\mu} + W_\mu^2 W^{2,\mu})$$

Back to ★, we get

$$\frac{1}{2} (D_\mu \Phi)^T (D^\mu \Phi) = \frac{1}{2} (\partial_\mu h) (\partial^\mu h) + \frac{g^2}{2} (v+h)^2 (W_\mu^1 W^{1,\mu} + W_\mu^2 W^{2,\mu})$$

We thus get

$$\begin{aligned} \frac{1}{2} (D_\mu \Phi)^T (D^\mu \Phi) &= \frac{1}{2} (\partial_\mu h) (\partial^\mu h) + \frac{1}{2} g^2 v^2 (W_\mu^1 W^{1,\mu} + W_\mu^2 W^{2,\mu}) \\ &\quad + \frac{g^2}{2} (h^2 + 2hv) (W_\mu^1 W^{1,\mu} + W_\mu^2 W^{2,\mu}) \end{aligned}$$

Added explanation (not in the original notes) — The ε - ε contraction, and reading the mass matrix off

The identity, step by step. With $(t^A)_{ab} = -i\varepsilon_{Aab}$,

$$(t^A t^B)_{ac} = (-i)^2 \varepsilon_{Aab} \varepsilon_{Bbc} = -\varepsilon_{Aab} \varepsilon_{Bbc}.$$

To use $\varepsilon_{ijk} \varepsilon_{lmk} = \delta_{il} \delta_{jm} - \delta_{im} \delta_{jl}$ the summed index b must sit last in both. It already does in ε_{Aab} ; and $\varepsilon_{Bbc} = \varepsilon_{cBb}$ by cyclicity. Hence

$$\varepsilon_{Aab} \varepsilon_{cBb} = \delta_{Ac} \delta_{aB} - \delta_{AB} \delta_{ac} \implies (t^A t^B)_{ac} = \delta_{AB} \delta_{ac} - \delta_{Ac} \delta_{aB},$$

which is the notes' second line. Sandwiching with $\Phi_0 = (0, 0, v+h)^T$ gives $\delta_{AB} (v+h)^2 - \delta_{A3} \delta_{B3} (v+h)^2$: the identity minus a projector onto the third direction, i.e. $\text{diag}(1, 1, 0)(v+h)^2$. ✓

The mass, and a convention warning. Comparing $\frac{1}{2} g^2 v^2 (W^1 W^1 + W^2 W^2)$ with the Proca normalisation $\frac{1}{2} m^2 W^\alpha W^\alpha$ gives

$$\boxed{m_{W^1} = m_{W^2} = g v}, \quad m_{W^3} = 0.$$

This is *not* the same numerical formula as $m_W = \frac{1}{2}gv$ in Example 1 — and that is correct, not an inconsistency. Three things differ: the scalar sits in the adjoint rather than the fundamental, the kinetic term carries an explicit $\frac{1}{2}$ (as it must for a real field), and the vacuum normalisation is $\Phi^T \Phi = v^2$ rather than $H^\dagger H = v^2/2$. The *structure* — mass $\propto g \times (\text{vev})$, only for broken directions — is what carries over.

Why W^1 and W^2 are degenerate here. Not by accident: they are exchanged by the unbroken $U(1)$ (see [p. 28*****]), so they must have equal mass. Degeneracy always signals a surviving symmetry.

Only two gauge bosons, W^1 and W^2 , get a mass (and interact with the Higgs) while W^3 remains massless [p. 27*****]

12 The unbroken $U(1)$, and electric charge

[p. 27*****]–[p. 29*****]

We now ask an important question: what is the action of the unbroken $U(1)$?

We consider the action of a global $U(1)$ transformation generated by t^3

$$\exp(-i\theta t^3) = \begin{pmatrix} \cos \theta & -\sin \theta & 0 \\ \sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

(written in the **adjoint rep**, since both Φ and the gauge bosons transform in this way)

on the different fields of the theory

■ SCALAR FIELDS (the Higgs)

We have

$$\Phi(x) = \begin{pmatrix} 0 \\ 0 \\ h+v \end{pmatrix} \longrightarrow \exp(-i\theta t^3) \begin{pmatrix} 0 \\ 0 \\ h+v \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ h+v \end{pmatrix}$$

In terms of the Higgs field:

$$h(x) \xrightarrow[U(1)]{\text{unbroken}} h(x)$$

$h(x)$ **does not transform** under the unbroken $U(1)$

Equivalently, we say that the Higgs field has zero $U(1)$ charge

$\hookrightarrow$ The particles created/destroyed by $h(x)$ have **zero charge**

[p. 28*****]

■ GAUGE BOSONS

Gauge bosons transform according to the adjoint of $SU(2)$. Restricted to the unbroken $U(1)$ generated by t^3 , we write

$$\begin{pmatrix} W_\mu^1 \\ W_\mu^2 \\ W_\mu^3 \end{pmatrix} \longrightarrow \left(\exp(-i\theta t^3) \right)_{AB} W_\mu^B = \begin{pmatrix} \cos \theta & -\sin \theta & 0 \\ \sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} W_\mu^1 \\ W_\mu^2 \\ W_\mu^3 \end{pmatrix} = \begin{pmatrix} \cos \theta W_\mu^1 - \sin \theta W_\mu^2 \\ \sin \theta W_\mu^1 + \cos \theta W_\mu^2 \\ W_\mu^3 \end{pmatrix}$$

Added explanation (not in the original notes) — slip flagged

In the original the rotation matrix is written with $+\sin \theta$ in the upper-right entry, which would contradict the result quoted immediately below it. The result $(\cos \theta W^1 - \sin \theta W^2$ in the first slot) is right and matches $\exp(-i\theta t^3)$ as boxed on [p. 27*****]; the matrix has been corrected above to agree.

We thus find

$$W_\mu^3 \xrightarrow[U(1)]{\text{unbroken}} W_\mu^3$$

W_μ^3 **does not transform** under unbroken $U(1)$
 $\rightsquigarrow$ it has **zero charge**

$$\begin{pmatrix} W_\mu^1 \\ W_\mu^2 \end{pmatrix} \xrightarrow[U(1)]{\text{unbroken}} \begin{pmatrix} \cos \theta W_\mu^1 - \sin \theta W_\mu^2 \\ \sin \theta W_\mu^1 + \cos \theta W_\mu^2 \end{pmatrix}$$

(W^1, W^2) transforms as a **vector** under $U(1)$ unbroken

It is useful to introduce the following **linear combinations**

[p. 29*****]

$$W_\mu^+ = \frac{1}{\sqrt{2}}(W_\mu^1 - iW_\mu^2)$$

$$W_\mu^- = \frac{1}{\sqrt{2}}(W_\mu^1 + iW_\mu^2)$$

Let us check how do they transform:

$$\begin{aligned} W_\mu^+ &= \frac{1}{\sqrt{2}}(W_\mu^1 - iW_\mu^2) \longrightarrow \frac{1}{\sqrt{2}}(\cos \theta W_\mu^1 - \sin \theta W_\mu^2 - i \sin \theta W_\mu^1 - i \cos \theta W_\mu^2) \\ &= \frac{1}{\sqrt{2}} \cos \theta (W_\mu^1 - iW_\mu^2) - \frac{i}{\sqrt{2}} \sin \theta (W_\mu^1 - iW_\mu^2) = \\ &= W_\mu^+ (\cos \theta - i \sin \theta) = e^{-i\theta} W_\mu^+ \\ W_\mu^- &= \frac{1}{\sqrt{2}}(W_\mu^1 + iW_\mu^2) \longrightarrow e^{+i\theta} W_\mu^- \end{aligned}$$

$$W_\mu^+ \xrightarrow[U(1)]{\text{unbroken}} e^{-i\theta} W_\mu^+$$

$$W_\mu^- \xrightarrow[U(1)]{\text{unbroken}} e^{+i\theta} W_\mu^-$$

W_μ^+ transforms according to the irrep of $U(1)$ with **charge +1**

W_μ^- transforms according to the irrep of $U(1)$ with **charge -1**

$W_\mu^\pm$ create/destroy massive spin-1 particles/antiparticles with charges ± 1 .

Added explanation (not in the original notes) — What has just been built — read this before moving on

Step back from the algebra. Starting from a pure $SU(2)$ gauge theory with an adjoint scalar, the theory has produced, entirely from the breaking pattern:

field	mass	$U(1)$ charge	interpretation
W_μ^3	0	0	a photon
$W_\mu^\pm$	gv	± 1	massive charged vectors
h	$\sqrt{2\lambda} v$	0	a neutral Higgs

This is a theory of electromagnetism plus charged massive vector bosons, obtained without ever postulating electromagnetism. The surviving gauge symmetry *is* the $U(1)$ of electrodynamics, its massless gauge field *is* the photon, and the charge under it is a label handed out by the unbroken generator. That is the content of the Georgi–Glashow model (1972), which was a serious competitor to the Weinberg–Salam theory until neutral currents were observed in 1973 — this model has no Z .

Why complexify W^1, W^2 ? Because a real field cannot carry a conserved charge. The $U(1)$ acts on (W^1, W^2) by rotation — a real two-dimensional representation — and the combinations diagonalising a rotation are always $W^1 \mp iW^2$, with eigenvalues $e^{\mp i\theta}$. Only in that basis is “charge” a well-defined quantum number. Exactly the same manoeuvre turns two real scalars into one complex charged scalar, and it is why W^+ and W^- are each other’s antiparticles: $(W_\mu^+)^* = W_\mu^-$.

The general lesson for the Standard Model. Everything about “which particle carries which charge” followed from one question: *how does the unbroken generator act?* On [p. 4*****] the same question is asked of $SU(2)_L \times U(1)_Y$, the unbroken generator turns out to be $Q = T^3 + Y$, and the answer is the electric charge assignment of the entire Standard Model.

Added explanation (not in the original notes) — PDF page 24 is blank

The back of sheet [p. 29*****]. Nothing is missing; the next sheet begins a new chapter.

PART V — A Standard Model primer [p. 1*****]–[p. 6*****]

Added explanation (not in the original notes) — New chapter — the star count changes again

From here the handwritten page numbers restart at 1, so the margin markers carry *fifteen* stars, [p. 1*****]–[p. 6*****], rather than the fourteen of [p. 29*****] and everything before it. Nothing is missing.

Read this chapter as the payoff. Everything in it is an application of machinery already built: the general theorem of Part II tells you how many gauge bosons get mass, and the Georgi–Glashow analysis of Part IV tells you how to find the unbroken generator and read off charges. The only genuinely new ingredient is that the gauge group is a *product*, $SU(2)_L \times U(1)_Y$, so the unbroken generator can be — and is — a *mixture* of generators from the two factors. That single fact is what produces the photon, the Z , and the weak mixing angle.

13 The gauge group and the bosonic sector

[p. 1*****]

A STANDARD MODEL PRIMER

The Standard Model is a chiral gauge theory based on the **gauge group**

[p. 1*****]

$$\begin{array}{ccc}
 SU(3)_c \times SU(2)_L \times U(1)_Y & & \\
 \underbrace{\hspace{10em}} & & \\
 \downarrow & \text{spontaneous} & \\
 U(1)_{\text{QED}} & \text{breaking} &
 \end{array}$$

■ THE BOSONIC SECTOR

The starting point is a **pure Yang–Mills theory** based on $SU(3) \times SU(2) \times U(1)$

$$\mathcal{L}_{\text{YM}} = -\frac{1}{4} \underbrace{F_{\mu\nu}^A F^{A,\mu\nu}}_{(1)} - \frac{1}{4} \underbrace{W_{\mu\nu}^A W^{A,\mu\nu}}_{(2)} - \frac{1}{4} \underbrace{B_{\mu\nu} B^{\mu\nu}}_{(3)}$$

$$\begin{array}{lll}
 (1) \quad F_{\mu\nu}^A = \partial_\mu G_\nu^A - \partial_\nu G_\mu^A - g_s f_{ABC} G_\mu^B G_\nu^C & (2) \quad W_{\mu\nu}^A = \partial_\mu W_\nu^A - \partial_\nu W_\mu^A - g \varepsilon_{ABC} W_\mu^B W_\nu^C & (3) \quad B_{\mu\nu} = \partial_\mu B_\nu - \partial_\nu B_\mu \\
 G_\mu^A, A = 1, \dots, 8 \quad \text{“gluons”} & W_\mu^A, A = 1, 2, 3 &
 \end{array}$$

To make some of the gauge bosons massive, the theory features a **spontaneous symmetry breaking**

Added explanation (not in the original notes) — Three field strengths, three different structures — why

The three terms look parallel but are not. The differences encode the entire structure of the Standard Model’s forces.

- $F_{\mu\nu}^A, SU(3)_c$: eight generators ($\dim SU(3) = 3^2 - 1 = 8$), structure constants f_{ABC} , coupling g_s .

Gluons self-interact. *Nothing in the scalar sector breaks $SU(3)_c$* — the Higgs is a colour singlet — so all eight gluons stay exactly massless. Colour confinement is a separate, non-perturbative story.

- $W_{\mu\nu}^A$, $SU(2)_L$: three generators, structure constants ε_{ABC} (this is the group of Parts III and IV), coupling g . The subscript L records that only left-handed fermions are charged under it — that is what “chiral gauge theory” means, and it is the one feature of the Standard Model that has nothing to do with this scan’s machinery.

- $B_{\mu\nu}$, $U(1)_Y$: *no* structure constants — abelian, so no self-interaction and no index on B_μ . Coupling g_Y (often written g').

One thing to notice already. W_μ^3 and B_μ are both electrically neutral, both massless before breaking, and both singlets of colour. Nothing forbids them from mixing once the symmetry is broken — and they will. That is the sole reason the Standard Model has *two* neutral massive/massless states (Z and γ) rather than the naive one-per-factor.

14 The Higgs doublet

[p. 2*****]

In the SM, the SSB is realised by introducing a **doublet of complex scalar fields**

[p. 2*****]

$$H(x) = \begin{pmatrix} H_1(x) \\ H_2(x) \end{pmatrix} \sim (1, 2, \frac{1}{2})$$

We need to specify **how H transforms under the gauge group**, and this notation means that

- H does not transform under $SU(3)_c$
- H transforms in the fundamental of $SU(2)_L$
- H transforms as the representation of $U(1)_Y$ with “hypercharge” $\frac{1}{2}$

Consequently

$$\begin{aligned} H(x) &\xrightarrow{SU(2)_L} \exp\left(-i\alpha_A(x)\frac{\sigma^A}{2}\right)H(x) \\ H(x) &\xrightarrow{U(1)_Y} \exp\left(-i\theta(x)\frac{1}{2}\mathbf{1}_{2\times 2}\right)H(x) \end{aligned}$$

We can also introduce a **covariant derivative**

$$D_\mu H = \partial_\mu H + ig W_\mu^A \frac{\sigma^A}{2} H + ig_Y B_\mu \frac{1}{2} H$$

Added explanation (not in the original notes) — How to read the label $(1, 2, \frac{1}{2})$

This compact notation is used for every field in the Standard Model, so it is worth learning properly. The three entries are the representation under each factor of $SU(3)_c \times SU(2)_L \times U(1)_Y$, in that order:

slot	value for H	meaning
$SU(3)_c$	1	singlet: H carries no colour, no coupling to gluons
$SU(2)_L$	2	doublet: the fundamental, generators $\sigma^A/2$
$U(1)_Y$	$\frac{1}{2}$	the hypercharge <i>number</i> Y , so $D_\mu \supset ig_Y B_\mu Y$

Why the abelian entry is a number, not a boldface symbol. All irreps of $U(1)$ are one-dimensional and labelled by a single charge. So there is no “dimension” to quote — only the charge itself. This is exactly the $W^\pm$ story of [p. 29*****]: a $U(1)$ irrep is a number.

Why the covariant derivative has that form. One term per gauge factor, each of the form $ig_{\text{factor}} \times (\text{gauge field}) \times (\text{generator in this rep})$. The $SU(3)$ piece is absent *because the generator is zero in the singlet rep* — not because we chose to leave it out. The $U(1)_Y$ generator is the number $\frac{1}{2}$ times the identity, which is why $\frac{1}{2}\mathbf{1}_{2\times 2}$ appears.

Why $Y = \frac{1}{2}$ and not something else. At this stage it is a choice; the physics only depends on it through $Q = T^3 + Y$ (see [p. 4*****]). Choosing $Y_H = \frac{1}{2}$ makes the lower component of H — the one that gets the vev — come out with $Q = -\frac{1}{2} + \frac{1}{2} = 0$, i.e. *electrically neutral*. That is

not optional: a charged vev would break electromagnetism and give the photon a mass. So $Y_H = \frac{1}{2}$ is fixed by the demand that the photon survive.

15 The bosonic Lagrangian, and the unbroken generator

[p. 3*****]-[p. 4*****]

We write the Lagrangian for the **bosonic sector** of the Standard Model

[p. 3*****]

$$\mathcal{L}_{\text{SM}}^{(B)} = -\frac{1}{4}F_{\mu\nu}^A F^{A,\mu\nu} - \frac{1}{4}W_{\mu\nu}^A W^{A,\mu\nu} - \frac{1}{4}B_{\mu\nu}B^{\mu\nu} \\ + (D_\mu H)^\dagger (D^\mu H) + \underbrace{\mu^2 H^\dagger H - \lambda (H^\dagger H)^2}_{\mu^2 > 0, \lambda > 0, \text{SSB}}$$

↪ Degenerate minima described by the condition

$$H^\dagger H = \frac{\mu^2}{2\lambda} = \frac{v^2}{2}$$

↪ We choose the vacuum

$$H_0 = \langle H \rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ v \end{pmatrix}$$

Formally, the SSB is standard. **The crucial point is to understand which symmetries, if any, are left unbroken.**

We consider the generators of $SU(2)_L \times U(1)_Y$ in the representation under which H transforms:

$$SU(2)_L : T^A \equiv \frac{\sigma^A}{2}, \quad A = 1, 2, 3 \quad ; \quad U(1)_Y : Y \equiv \frac{1}{2} \mathbf{1}_{2 \times 2}$$

We have that

$$T^A H_0 \neq 0, \quad A = 1, 2, 3 \quad \text{and} \quad Y H_0 \neq 0$$

[p. 4*****]

However, if we consider the **combination**

$$Q = T^3 + Y$$

↪

$$Q H_0 = 0$$

$$Q H_0 = \frac{1}{2} \left[\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} + \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \right] \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ v \end{pmatrix} = \\ = \frac{1}{2} \begin{pmatrix} 2 & 0 \\ 0 & 0 \end{pmatrix} \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ v \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ v \end{pmatrix} = 0$$

The hermitian generator $Q = T^3 + Y$ annihilates the vacuum

↪ The vacuum is invariant under the symmetry generated by Q

Since it only has one hermitian generator, such group is $U(1)$. We call it $U(1)_{\text{em}}$

We thus have the breaking pattern

$$SU(2)_L \times U(1)_Y \longrightarrow U(1)_{\text{em}}$$

which delivers 3 Goldstone bosons

Added explanation (not in the original notes) — The single most important calculation in the Standard Model, unpacked

Why no individual generator works, but a combination does. Each of T^1, T^2, T^3, Y fails on its own:

$$T^1 H_0 \propto \begin{pmatrix} v \\ 0 \end{pmatrix}, \quad T^2 H_0 \propto \begin{pmatrix} -iv \\ 0 \end{pmatrix}, \quad T^3 H_0 = \frac{1}{2\sqrt{2}} \begin{pmatrix} 0 \\ -v \end{pmatrix}, \quad Y H_0 = \frac{1}{2\sqrt{2}} \begin{pmatrix} 0 \\ v \end{pmatrix}.$$

Look at the last two: they are *equal and opposite*. Their sum cancels exactly. T^1, T^2 point in the “upper” direction and cannot be cancelled by anything — there is no other generator producing an upper component. So $T^3 + Y$ is the unique surviving direction, up to normalisation, and $4 - 1 = 3$ generators are broken, giving 3 Goldstones.

Why this is only possible for a product group. In Example 1, $G = SU(2)$ alone, and no combination of $T^{1,2,3}$ annihilated H_0 — everything broke. Adding the $U(1)_Y$ factor supplies exactly the extra generator needed to cancel T^3 . *Electromagnetism exists because the electroweak group is a product.* The photon is not the $U(1)_Y$ gauge boson and it is not W^3 ; it is a specific mixture, and the mixing angle is a free parameter of the theory (measured, not predicted).

Gell-Mann–Nishijima. The relation $Q = T^3 + Y$ is exactly the formula relating electric charge, weak isospin and hypercharge that was known empirically from hadron spectroscopy long before the electroweak theory. Here it is not fitted — it *drops out* of asking which generator annihilates the vacuum. Checking it on H itself: the upper component has $T^3 = +\frac{1}{2}$, $Y = +\frac{1}{2}$, hence $Q = +1$; the lower has $T^3 = -\frac{1}{2}$, $Y = +\frac{1}{2}$, hence $Q = 0$. So $H = (H^+, H^0)^T$, and it is the *neutral* component that condenses. ✓

The counting, before doing any more work. $\dim(SU(2)_L \times U(1)_Y) - \dim U(1)_{\text{em}} = (3+1) - 1 = 3$. By the theorem of [p. 13*****]: three gauge bosons become massive, one stays exactly massless. Those are $W^\pm$, Z , and γ . The Higgs doublet has 4 real components; 3 are eaten, leaving 1 physical scalar — the Higgs boson. *The entire electroweak spectrum has been derived before writing down a single mass formula.*

16 The mass spectrum: the Higgs field

[p. 5*****]

In the unitary gauge, the 3 Goldstone bosons are **gauged away** by the gauge fixing along the broken directions

[p. 5*****]

$$H(x) = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ v + h(x) \end{pmatrix}$$

UNITARY GAUGE

We now discuss the mass spectrum of the theory

■ THE HIGGS FIELD

From the potential, we get

$$\begin{aligned} V(H^\dagger H) &= -\mu^2 H^\dagger H + \lambda (H^\dagger H)^2 \\ \downarrow \quad H &= \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ v + h \end{pmatrix} \\ V(h) &= \frac{1}{2} (2\mu^2) h^2 + \lambda v h^3 + \frac{\lambda}{4} h^4 \end{aligned}$$

The Higgs field h is **massive**, with mass

$$m_h^2 = 2\mu^2 = 2\lambda v^2$$

17 The gauge bosons: the quadratic terms

[p. 6*****]

■ THE GAUGE BOSONS

We investigate the kinetic term of the Higgs field, $\underbrace{(D_\mu H)^\dagger (D^\mu H)}_{(*)}$

[p. 6*****]

$$D^\mu H = \partial^\mu H + ig W^{A,\mu} \frac{\sigma^A}{2} H + i \frac{g_Y}{2} B^\mu \mathbf{1} H$$

$$(D_\mu H)^\dagger = \partial_\mu H^\dagger - ig W_\mu^A H^\dagger \frac{\sigma^A}{2} - i \frac{g_Y}{2} B_\mu H^\dagger$$

$$(*) = \left[\partial_\mu H^\dagger - ig W_\mu^A H^\dagger \frac{\sigma^A}{2} - i \frac{g_Y}{2} B_\mu H^\dagger \right] \left[\partial^\mu H + ig W^{A,\mu} \frac{\sigma^A}{2} H + i \frac{g_Y}{2} B^\mu H \right] =$$

$$= \frac{1}{2} (\partial_\mu h) (\partial^\mu h) + \frac{g^2}{8} (h+v)^2 \delta_{AB} W_\mu^A W^{B,\mu} + \frac{g_Y^2}{8} (v+h)^2 B_\mu B^\mu$$

$$+ \frac{g g_Y}{4} W_\mu^A B^\mu \underbrace{H^\dagger \sigma^A H}_{(\dagger)} \times 2 =$$

$$= \underbrace{\frac{1}{2} (\partial_\mu h) (\partial^\mu h)}_{\text{conventional kinetic term for the Higgs}} + \underbrace{\frac{g^2}{8} (h+v)^2 \delta_{AB} W_\mu^A W^{B,\mu} + \frac{g_Y^2}{8} (v+h)^2 B_\mu B^\mu - \frac{g g_Y}{8} W_\mu^3 B^\mu (h+v)^2 \cdot 2}_{\text{we consider these terms, which are quadratic in the gauge fields}}$$

$$(\dagger) \quad \frac{1}{2}(0, h+v)\sigma^A \begin{pmatrix} 0 \\ h+v \end{pmatrix} = \delta_{A3} \frac{1}{2}(-1)(h+v)^2$$

[the scan ends here]

Added explanation (not in the original notes) — What is different this time — and why the scan stopping here is frustrating

Compare the three lines above with the corresponding computation in Example 1 ([p. 18*****]). Two of the three terms are familiar:

- $\frac{g^2}{8}(v+h)^2\delta_{AB}W^AW^B$ — identical in form to Example 1, and coming from the same Pauli identity $\sigma^A\sigma^B = \delta_{AB}\mathbf{1} + i\varepsilon_{ABC}\sigma^C$ with the ε piece killed by symmetry.
- $\frac{g_Y^2}{8}(v+h)^2B_\mu B^\mu$ — new but trivial: the $U(1)_Y$ generator is a number, so there is no index structure at all.

The third term is the one that changes everything:

$$-\frac{g g_Y}{4}(v+h)^2 W_\mu^3 B^\mu.$$

This is an off-diagonal mass term. It says W^3 and B are *not* mass eigenstates. Unlike every previous example in this scan — where the mass matrix came out proportional to δ_{AB} and the fields we started with were already the physical particles — here one must diagonalise a genuine 2×2 matrix before the spectrum can be read off.

Why the cross term exists at all, in one line. It requires a generator that is non-zero on the vacuum in *both* factors — i.e. $H_0^\dagger \sigma^3 H_0 \neq 0$ and $Y \neq 0$. Both hold. And its coefficient is exactly $-2 \times$ the geometric mean of the two diagonal ones, which (as the next section shows) forces one eigenvalue to be *exactly zero*. That zero is the photon, and it is guaranteed by the $QH_0 = 0$ of [p. 4*****].

The notes break off precisely here, one step short. Section 18 takes it.

18 Where this is going — diagonalising the neutral sector

Added explanation (not in the original notes) — Everything in this section is added; it is not in the original notes

The scan stops with the quadratic terms written but not diagonalised. What follows is the rest of the calculation — three short steps — because the result is the electroweak spectrum, and because it is the natural end of the arc that began in Lecture 24.

Step 1 — collect the mass terms

Set $\hbar \rightarrow 0$ in the quadratic terms of [p. 6*****] (the $\hbar$ -dependent pieces are interactions, not masses):

$$\mathcal{L}_{\text{mass}} = \frac{v^2}{8} \left[g^2 (W_\mu^1 W^{1,\mu} + W_\mu^2 W^{2,\mu}) + g^2 W_\mu^3 W^{3,\mu} - 2g g_Y W_\mu^3 B^\mu + g_Y^2 B_\mu B^\mu \right].$$

The charged sector is already diagonal. Defining, exactly as in the Georgi–Glashow model [p. 29*****],

$$W_\mu^\pm = \frac{1}{\sqrt{2}} (W_\mu^1 \mp i W_\mu^2), \quad W^1 W^1 + W^2 W^2 = 2 W_\mu^+ W^{-,\mu},$$

and comparing with the Proca normalisation $m_W^2 W_\mu^+ W^{-,\mu}$ for a complex vector:

$$m_W = \frac{g v}{2}$$

— identical to Example 1 of [p. 19*****], as it must be, since the charged sector never noticed the extra $U(1)_Y$.

Step 2 — diagonalise the neutral 2×2 block

The neutral part is a perfect square:

$$\frac{v^2}{8} \left[g^2 W^3 W^3 - 2g g_Y W^3 B + g_Y^2 B B \right] = \frac{v^2}{8} \left(g W_\mu^3 - g_Y B_\mu \right)^2.$$

Written as $\frac{1}{2} \begin{pmatrix} W^3 & B \end{pmatrix} M^2 \begin{pmatrix} W^3 \\ B \end{pmatrix}$,

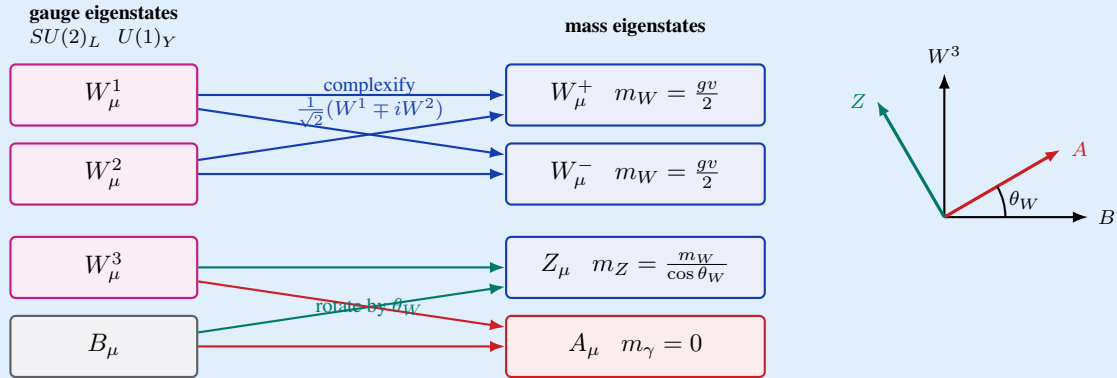
$$M^2 = \frac{v^2}{4} \begin{pmatrix} g^2 & -g g_Y \\ -g g_Y & g_Y^2 \end{pmatrix}, \quad \det M^2 = \frac{v^4}{16} (g^2 g_Y^2 - g^2 g_Y^2) = 0, \quad \text{tr } M^2 = \frac{v^2}{4} (g^2 + g_Y^2).$$

A vanishing determinant means **one eigenvalue is exactly zero**. The eigenvalues are therefore 0 and $\frac{v^2}{4} (g^2 + g_Y^2)$. Defining the **weak mixing angle** by $\tan \theta_W \equiv g_Y/g$, the normalised eigenvectors are

$$\begin{aligned} A_\mu &= \sin \theta_W W_\mu^3 + \cos \theta_W B_\mu & \text{eigenvalue } 0 &\implies m_\gamma = 0 \\ Z_\mu &= \cos \theta_W W_\mu^3 - \sin \theta_W B_\mu & \text{eigenvalue } \frac{v^2}{4} (g^2 + g_Y^2) &\implies m_Z = \frac{v}{2} \sqrt{g^2 + g_Y^2} \end{aligned}$$

This is an orthogonal rotation of (W^3, B) by the angle θ_W , so the kinetic terms stay canonical.

Added explanation (not in the original notes) — Figure — the electroweak rotation, and where each particle comes from



Measured values: $\sin^2 \theta_W \simeq 0.231$, so $\theta_W \simeq 28.7^\circ$. The photon is mostly B and partly W^3 ; the Z is mostly W^3 and partly B . Neither is “the” gauge boson of either factor — which is exactly why neither g nor g_Y alone is the electric charge.

Step 3 — the consequences

The photon is massless, and its coupling is the electric charge. Substituting the inverse rotation into D_μ , the coefficient of A_μ acting on a state of charge Q is

$$e = g \sin \theta_W = g_Y \cos \theta_W = \frac{g g_Y}{\sqrt{g^2 + g_Y^2}},$$

which is symmetric in the two couplings — as it must be, since the photon does not know which factor it came from. Note $e < g$ and $e < g_Y$: the electromagnetic coupling is weaker than either of its parents.

The ρ parameter. From $m_W = \frac{gv}{2}$ and $m_Z = \frac{v}{2} \sqrt{g^2 + g_Y^2} = \frac{gv}{2 \cos \theta_W}$,

$$m_Z = \frac{m_W}{\cos \theta_W} \iff \rho \equiv \frac{m_W^2}{m_Z^2 \cos^2 \theta_W} = 1.$$

This is a genuine *prediction*, and a sharp one: three measured quantities constrained by one relation, with no free parameter left to absorb a discrepancy. Experimentally $\rho = 1.00038 \pm 0.00020$.

Added explanation (not in the original notes) — Why $\rho = 1$ is not an accident — custodial symmetry

$\rho = 1$ traces back to the same custodial $SU(2)$ mentioned in Part III. The Higgs potential depends on H only through $H^\dagger H$, which is invariant under the full $SO(4) \cong SU(2)_L \times SU(2)_R$ acting on the four real components of the doublet. The vev breaks this to the diagonal $SU(2)_{\text{custodial}}$, and the three would-be Goldstones form a *triplet* of it — hence the three gauge bosons that eat them are degenerate in the limit $g_Y \rightarrow 0$. Switching g_Y back on splits W^3 from $W^{1,2}$ by precisely the factor $\cos \theta_W$, which is what $\rho = 1$ records.

Why this matters. $\rho = 1$ is a consequence of the Higgs being a *doublet*. A scalar in a different representation (a triplet, say) would generically give $\rho \neq 1$ at tree level. The measured value is therefore strong evidence that the field responsible for electroweak breaking really is an $SU(2)_L$ doublet — which is not something the theory forced on us; it was a choice on [p. 2*****], and experiment confirms it.

Custodial symmetry is only approximate. It is broken by g_Y (i.e. by hypercharge) and by the Yukawa couplings, chiefly the large top–bottom mass splitting. Those effects appear as small loop corrections to ρ — and their smallness is what makes the measurement above so constraining.

Added explanation (not in the original notes) — The finished electroweak spectrum, and how many inputs it took

particle	mass	Q	origin	measured
$W^\pm$	$\frac{1}{2}gv$	± 1	eats $\pi^\pm$	80.4 GeV
Z	$\frac{1}{2}\sqrt{g^2 + g_Y^2}v$	0	eats π^0	91.2 GeV
γ	0	0	unbroken $Q = T^3 + Y$	0
h	$\sqrt{2\lambda}v$	0	the surviving scalar	125 GeV
$g^A (\times 8)$	0	0	$SU(3)_c$ untouched	0

Input counting. The bosonic sector has four parameters: g, g_Y, μ^2, λ (plus g_s , which decouples from all of this). Equivalently: e, θ_W, v, m_h . From these, *everything* above is predicted — four inputs, five masses plus every coupling among them. The one leftover relation is $\rho = 1$.

And v is measured how? From the muon lifetime: $G_F/\sqrt{2} = 1/(2v^2)$, giving $v = 246$ GeV. So the scale of electroweak symmetry breaking was known, to good accuracy, decades before the Higgs boson was found.

Summary of Lecture 30 — the logic in twelve lines

Added explanation (not in the original notes) — Summary

Part I — closing the abelian chapter

1. Substituting $\varepsilon_L^\mu \simeq p^\mu/gv$ into the $A_L A_L$ amplitude of Lecture 29, the gauge coupling cancels completely and $i\mathcal{M} \simeq -\frac{i}{v^2} \left[\frac{s^2}{s-m_h^2} + \frac{t^2}{t-m_h^2} + \frac{u^2}{u-m_h^2} \right]$ — *precisely* the four-Goldstone amplitude. The Goldstone equivalence theorem, verified. [p. 9*****]

Part II — the general non-abelian mechanism

2. Gauge a global $SU(N)$ with an SSB potential; evaluate $\mathcal{L}$ on the vacuum. Only $|D_\mu \Phi_0|^2$ survives, and it is quadratic in the gauge fields with no derivatives — a mass term. [p. 10*****]–[p. 12*****]
3. Unbroken generators annihilate Φ_0 , so only broken indices survive: $M_{\alpha\beta}^2 = 2g^2 \Phi_0^\dagger X^\alpha X^\beta \Phi_0$. [p. 12*****]
4. This is a **Gram matrix** of the vectors $X^\alpha \Phi_0$: hermitian, and positive definite because those vectors are linearly independent. Hence *every* broken direction gets a strictly positive mass, and $\dim G - \dim H$ **gauge bosons become massive** while $\dim H$ stay exactly massless. [p. 13*****]–[p. 14*****]

Part III — Example 1: $SU(2) \rightarrow 1$

5. A complex doublet with $\langle H \rangle = \frac{1}{\sqrt{2}}(0, v)^T$: all three $\sigma^A/2$ fail to annihilate it, so all three generators break, three Goldstones appear, $\mathcal{M}_0 \cong S^3$. [p. 14*****]–[p. 15*****]
6. Unitary gauge $\alpha_A = \pi^A/v$ removes all three; the Pauli identity kills the ε_{ABC} piece by symmetry, leaving a mass matrix $\propto \delta_{AB}$: three *degenerate* W 's with $m_W^2 = \frac{1}{4}g^2v^2$, plus $m_h^2 = 2\mu^2$. [p. 16*****]–[p. 19*****]
7. $W_L W_L$ scattering from the gauge diagrams alone grows as $g^2 E^2/m_W^2$, violating unitarity at $E \sim m_W/g \sim v$. The Higgs-exchange diagrams — with vertex $2m_W^2/v$, fixed by the same g and v — cancel that growth exactly. “*The Higgs unitarises the longitudinal WW scattering*”: it is the UV completion of massive Yang–Mills. [p. 19*****]–[p. 21*****]

Part IV — Example 2: Georgi–Glashow, $SU(2) \rightarrow U(1)$

8. Put the scalar in the **adjoint** instead: a real triplet, $(t^A)_{BC} = -i\varepsilon_{ABC}$, $\Phi^T \Phi$ invariant, $\mathcal{M}_0 \cong S^2$. With $\Phi_0 = (0, 0, v)^T$, t^1 and t^2 break but t^3 annihilates: $SU(2) \rightarrow SO(2) \cong U(1)$, two Goldstones. [p. 22*****]–[p. 24*****]
9. The $\varepsilon\varepsilon$ contraction gives $\Phi^T t^A t^B \Phi = (v+h)^2 \text{diag}(1, 1, 0)$: $W^{1,2}$ get mass gv , W^3 stays massless. [p. 25*****]–[p. 26*****]
10. Asking how the unbroken $U(1)$ acts: h and W^3 are inert (charge 0), while (W^1, W^2) rotate. Diagonalising, $W^\pm = \frac{1}{\sqrt{2}}(W^1 \mp iW^2)$ carry charge ± 1 . A photon and charged massive vectors, derived rather than postulated. [p. 27*****]–[p. 29*****]

Part V — the Standard Model

11. $SU(3)_c \times SU(2)_L \times U(1)_Y$ with a doublet $H \sim (1, 2, \frac{1}{2})$. Individually $T^{1,2,3}$ and Y all fail to annihilate H_0 — but $T^3 H_0$ and $Y H_0$ are equal and opposite, so $Q = T^3 + Y$ survives. Hence $SU(2)_L \times U(1)_Y \rightarrow U(1)_{\text{em}}$, three Goldstones, and the Gell-Mann–Nishijima relation falls out. [p. 1*****]–[p. 4*****]
12. In unitary gauge $m_h^2 = 2\lambda v^2$, and the quadratic gauge terms contain a $W_\mu^3 B^\mu$ *cross term* — new relative to every earlier example. [p. 5*****]–[p. 6*****] (*Diagonalised in*

Section 18: $m_W = \frac{1}{2}gv$, a massless photon, $m_Z = m_W / \cos \theta_W$, and $\rho = 1$.)

Added explanation (not in the original notes) — Where this is going next

The chapter title says *primer*, and only the bosonic sector has been built. Two things remain.

The fermions, and the word “chiral”. [p. 1*****] calls the Standard Model a *chiral* gauge theory and then never uses the word again. It means that left- and right-handed fermions sit in *different* representations — $Q_L \sim (\mathbf{3}, \mathbf{2}, \frac{1}{6})$ but $u_R \sim (\mathbf{3}, \mathbf{1}, \frac{2}{3})$, and so on. The immediate consequence is that a Dirac mass term $\bar{\psi}_L \psi_R$ is *forbidden* by gauge invariance: every fermion in the Standard Model would be massless were it not for the Higgs. Their masses come from Yukawa couplings $y \bar{Q}_L H d_R + \text{h.c.}$, which become $yv/\sqrt{2}$ after breaking — the same v again. So the Higgs is responsible for *all* the masses in the theory, not just the gauge bosons’.

Anomalies. A chiral gauge theory is not automatically consistent: gauge anomalies can destroy it. The hypercharge assignments of the Standard Model satisfy a set of highly non-trivial cancellation conditions — and they only work when quarks and leptons are counted *together*, generation by generation, with the colour factor of 3 doing essential work. This is the deepest structural hint in the Standard Model, and the usual entry point to grand unification.